Summary of Water Quality and Biological Data for Anadromous Reaches of the Jordan River

Prepared For:

BC Hydro
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EXECUTIVE SUMMARY

In the Jordan River Watershed Plan, re-introduction of pink salmon and restoration of habitat for multi species were identified as key priorities for the anadromous reaches of the river. However, before these initiatives can commence, it is important to determine whether copper toxicity, which has historically been an issue in the lower river, has diminished to a level where the river can support healthy fish populations during their various freshwater life stages. The purpose of this report was to address this question by pulling together available water chemistry data on Jordan River, as well as findings from various fisheries investigations on the river.

The various studies conducted on the river provided water quality and/or biological data spanning the years 1994 to 2012 and encompassed periods before and after initiation of the fish flow release from Elliott Dam in January 2008. The biological studies provided pertinent information on salmonid spawning, incubation, and rearing.

The various water samples collected on the anadromous reaches of the Jordan River show an abrupt increase in copper concentration beginning adjacent to the large deposit of material left on the left bank by historic mining activities. Dissolved copper data from within this impacted zone ranged from 0.007 – 0.086 mg/L, were inversely related to stream discharge, and fit well to a power curve equation. In contrast, copper levels upstream of the mine deposit ranged from <0.001 to 0.004 mg/L and did not vary with stream discharge.

Most studies conducted on the lower Jordan River have assumed that the mine deposit on the left bank is composed of copper "tailings", and that this is the source of elevated copper in the lower river. However, the inspection of this deposit by project engineer indicated that the particle size and colour are inconsistent with tailings, and that the deposit may in fact be excavations from creation of the access tunnel in 1956. Thus, there remains uncertainty as to the actual source of the copper.

Water quality monitoring by the one of the WUP monitoring studies (JORMON#2) found that copper levels within the impacted zone decreased incrementally, first in 2007, and then again in 2008. The first decrease was believed to be due record floods during the 2006/07 winter, which washed some of the mine deposit out to the ocean. The second decrease occurred following initiation of fish flow releases at Elliott Dam in January 2008. This decrease was due to the dilution effect provided by these flow releases. Due to concerns of the fish release control valve tripping, the valve has been maintained in a locked open position. In this position, flow releases at Elliott Dam typically range from 0.30 – 0.40 m^3/s as opposed to 0.25 m^3/s prescribed in the Jordan River WUP.

Spawning surveys (JORMON#3) were conducted from fall 2005 to spring 2011 and involved snorkel enumerations during coho and steelhead spawning periods. Numbers of adults counted during the 6 survey years were very low, but when fish were found, it was frequently in the riffles adjacent to the mine deposit. This suggests that adults did not avoid the copper impacted zone. Of note is that flows during spawning periods tend to be higher than in the summer, which would have diluted copper inputs.

Incubations studies using coho or pink salmon eggs were conducted from 2000 – 2011. Results generally indicated egg-to-emergence survival rates of ≥ 90%, with no apparent difference between sites within the impacted zone compared with those above this zone. Sites with poor survival were attributed to siltation, and/or low intergravel flow/dissolve oxygen conditions as opposed to copper toxicity. These studies suggest that incubation success is not diminished by ambient levels of copper within the impacted zone.

Results from rearing studies (total removal electrofishing) indicated a total absence of rearing fish within the impacted zone in 1994, 2005, and 2006. In 2007, rainbow fry were encountered for the first time, albeit fish displayed extremely low condition factors. The presence of fry in this zone in 2007 was attributed to a reduction in copper due to the previous winter's floods as previously described. However, the thinness of the fry suggested copper levels were still causing adverse effects on fish health. In January 2008, the fish flow commenced, and subsequent inventories from 2008 to 2010 captured both rainbow fry and parr within the impacted zone, and these were at densities similar to a control site upstream of the copper zone. In addition, fish captured in the copper affected zone during 2008 – 2010 showed condition factors similar to fish at the control site. The occurrence of multiple age classes, of similar densities and condition factors to the control site suggests that copper levels may be sufficiently reduced under the fish flow release to permit successful rearing of rainbow trout.

The performance of rearing prior 2007, in 2007, and after 2007, provided an opportunity to assess the threshold level of copper below which rearing becomes successful in the impacted zone. This was done by constructing a copper-discharge regression equation and applying this against mean monthly flows to estimate mean concentration of dissolved copper during rearing months for 2006 to 2010. Results suggested that rearing is successful when dissolved copper concentrations are less than ~ 0.015 – 0.020 mg/L. From this level to about 0.040 mg/L, rearing is possible but is impaired (sublethal impacts such as in 2007). When dissolved copper levels exceed 0.40 mg/L the data suggest rearing is not possible.

In conclusion, available water chemistry and biological data on the Jordan River suggest that, under the current flow releases from Elliott Dam (0.3 – 0.4 m^3/s), copper inputs into the lower river appear to be sufficiently diluted to allow for successful completion of spawning, incubation, and rearing. The data indicate that the rearing stage is the most sensitive to copper, but if concentrations remain below ~0.015 – 0.020 mg/L rearing can occur and fish are able to feed and grow normally (as indicated by condition factor in 2008 2010). The one uncertainty in this conclusion is whether there are more subtle adverse effects occurring that are not detectable by available Jordan River studies. Examples of subtle effects include impaired olfaction (sense of smell) and physiological stress responses.

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1. INTRODUCTION

In the Jordan River Watershed Plan (FWCP 2011), re-introduction of pink salmon and restoration of habitat for multi species were identified as key priorities for the lower Jordan River. However, before these initiatives can commence, the FWCP Coastal Board determined that it was important to first determine whether the water quality in the lower river is capable of supporting healthy fish populations during the various stages of freshwater development. The concern over water quality was due historic mine activities, which left a deposit of material along the left bank of the lower river that is believed to be leaching copper into the river. What needs to be determined is whether copper concentrations in this section of river allow for successful completion of the various salmonid freshwater life history stages. In support of this goal, a number of studies have recently been completed that provide both water quality and biological data, and potentially give insight into the capability of the lower river to sustain salmonid populations (WUP monitoring projects JORMON#2, JORMON#3 and FWCP Coastal Projects 02.Jo.64 and 04.Jo.01).

The objective of this report are to assist with the decision of whether it is appropriate to move forward with FWCP initiatives for the lower Jordan River by:

1. Compiling existing water quality and relevant biological data for the lower Jordan River.
2. Summarizing the biological data in terms of the status (success or failure) of salmonid life stages under copper levels occurring in the lower Jordan River.
3. If existing information is insufficient to answer the above question, then outline a water quality and/or biological program that fills these data gaps.

2. Methods

Water quality and fish biology data for the lower Jordan River were available from a number of studies conducted during 1994 to 2010. These reports were obtained from the following sources:

- BC Hydro— BC Hydro supplied the 1994 biophysical and fish production assessment by Griffith (1996), as well as Fish and Wildlife Compensation Program documents: pink salmon incubation study by Wright and Guimond (2003) and the pink salmon transplant feasibility study by Guimond (2005).
- DFO— DFO provided a memo report presenting the results from the Jordan River coho salmon bioassay study conducted from November 2000 to May 2001 (Lightly 2001).
- WUP monitoring programs— These include the Lower Jordan River Fish Index Study (JORMON#2) and the Lower Jordan River Salmon Spawning Assessment and Enumeration (JORMON#3). The first was available in-house since it was conducted by B. Burt and

Associates, while reports JORMON#3 were obtained from BC Hydro's website (http://www.bchydro.com/about/sustainability/conservation/water_use_planning/vancouver_island.html)

Water quality data from these reports were compiled into an Excel database. The main focus of the database was to collate copper records for various sample locations on the lower river, however, other water quality parameters were also input into the database. Since there appears to be a relationship between copper concentration and flow, stream discharge data for the lower river were added to the database. When possible, flow data were obtained from the Lower Jordan River Inflow Monitoring Program, which was a component of the Jordan River WUP Monitoring (JORMON#1) conducted by Hydroid Geoscience and D. Burt and Associates from 2005 to 2011. For copper records outside of this time period, flow data were calculated from BC Hydro mean daily total system inflows (TSI's) into Elliott Headpond, multiplied by the lower river/TSI watershed area ratio, and then adjusted with a regression correction factor based on the relationship between modelled and measured flows (details in Burt and Hudson 2012).

3. BACKGROUND

3.1 Setting

The Jordan River is located on the southwest coast of Vancouver Island 72 km by road from the city of Victoria. The river originates in the Seymour Mountain Range in south central Vancouver Island and flows in a south-westerly direction before emptying into Juan De Fuca Strait adjacent to the community of Jordan River. It has a drainage area of 162 km^2 and a mean annual discharge at the mouth of approximately 13.7 m^3/s (Cascadia Biological Services 2001).

In terms of the Provincial Ecoregion Classification System (Demarchi 1996), the Jordan Watershed spans two distinct ecosections. The lower Jordan River and western tributaries of the upper river lie in the Windward Island Mountains (WIM) Ecosection (Coast and Mountain Ecoprovince), while the mainstem of the upper river and its eastern tributaries lie within the Leeward Island Mountains (LIM) Ecosection (Georgia Depression Ecoprovince). These two ecosections differ in climate and geology which in turn results in differences in water chemistry. This may be of relevance after January 2008 when BC Hydro commenced fish flow releases from Elliott Dam bringing waters from the LIM Ecosection to the lower river which was previously dominated by WIM water chemistry.

3.2 BC Hydro Facilities

The Jordan River hydroelectric project was completed in 1911 and rebuilt in 1971. Current facilities include three dams, two reservoirs, a headpond, a tunnel/penstock water delivery system, and a 175 MW powerhouse on the lower Jordan River (BC Hydro 2003). The dams include Bear Creek Dam which impounds Bear Creek Reservoir, Jordan Diversion Dam which impounds

Diversion Reservoir, and Elliott Dam which impounds Elliott Headpond (Figure 1). The powerhouse is located on the west side of the Jordan River 900 m upstream of the mouth. This powerhouse replaced an older facility and tailrace in 1971. The original facility was located on the east side of the river and generation flows discharged into a 500 m long tailrace which emptied into the Jordan River estuary (Wright and Guimond 2003).

Prior to 2008, no flows were released from Elliott Dam except when spilling was required during flood events. As a result, flows in the river below Elliott Dam were generally derived entirely from local tributary inputs. This resulted in discharges as low as 10 – 12 L/s (0.010 – 0.012 m^3/s) immediately upstream of the generating station during July and August periods (Burt and Hudson 2008, 2009). In winter 2007/2008 Elliott Dam was outfitted with a fishwater release pipe and in January 2008 fish flow releases commenced at Elliott Dam. The agreement in the Jordan River WUP was to maintain a year-round release of 0.25 m^3/s via a flow control valve in the release pipe. However, due to concerns over the control valve tripping (which would result in valve closure and subsequent loss of fish flows) the valve has been maintained in a locked open position since its installation. In this configuration, flow releases at the dam have generally slowly cycled between 0.30 and 0.40 m^3/s as Elliott Headpond is drawn down and then refilled from Diversion Reservoir.

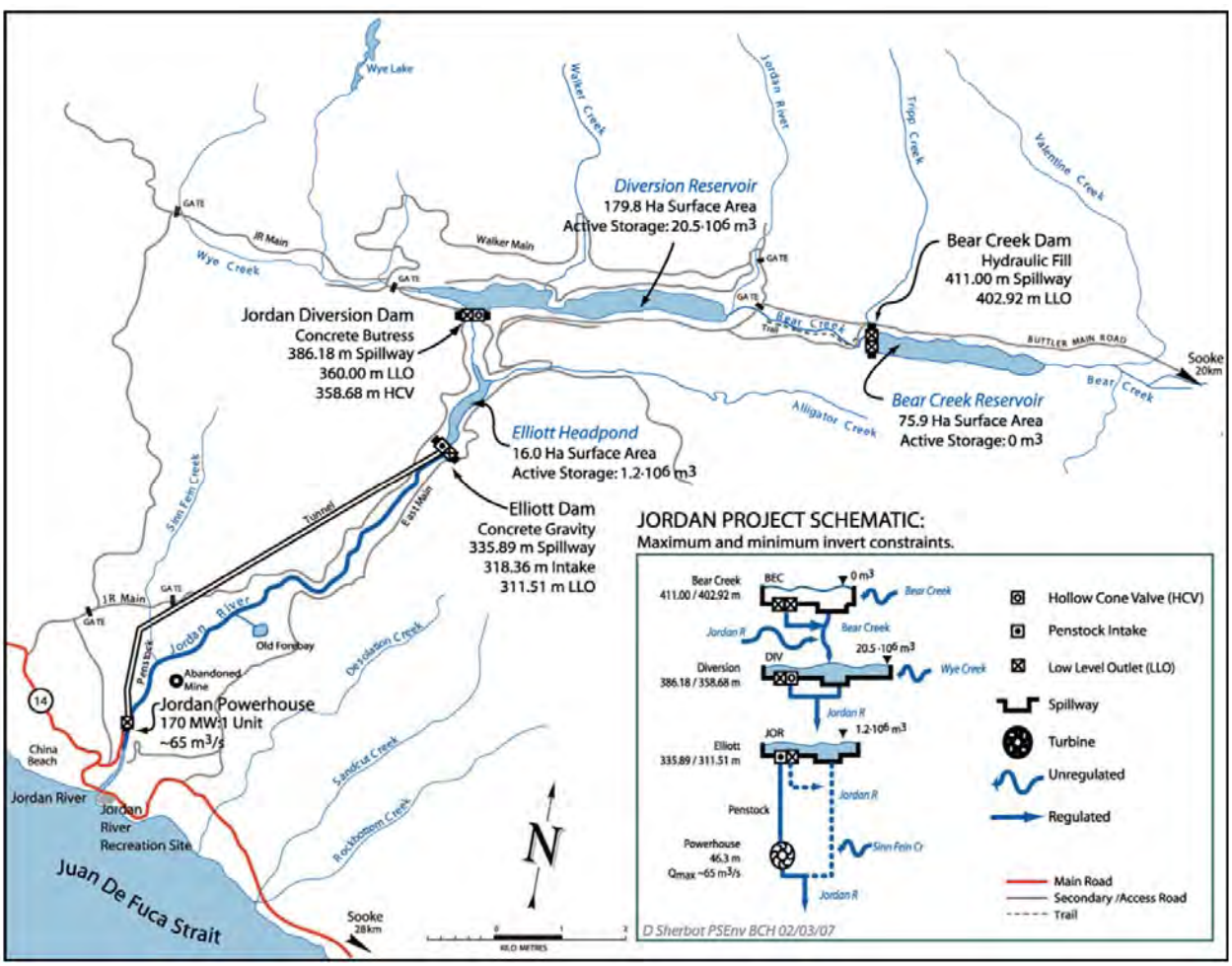

Figure 1. Map of the Jordan River Watershed showing BC Hydro facilities (from BC Hydro 2003).

3.3 Forestry

Logging operations in the Jordan Watershed date back to the 1880's (Wright and Guimond 2003). Beginning in 1934, forest harvest activities in the watershed were managed by Western Forest Products (WFP) and included a dryland sort built on land reclaimed from the estuary (built in the 1960's) and booming in the lower river. In 2008 WFP sold off private portions of the land base and in 2010 Crown portions were formed into their own Tree Farm Licence (TFL 65, 20,213 ha.). In 2011 the rights to TFL 65 were sold to the Pacheedaht Andersen Timber Holdings Limited Partnership (PATH), a partnership between the Pacheedaht First Nation of Port Renfrew and Andersen Timber of Vancouver. Day to day activities within the TFL are managed by their general partner, Queesto Community Forest Ltd. These operations will continue to use the existing dryland sort and booming will still be conduct in the lower river. In addition, dredging is required about every 2 years to maintain booming operations

(references: http://www.sookenewsmirror.com/news/118019999.html?mobile=true; http://queesto.ca/)

3.4 Mining Operations

Mining of copper and gold has occurred periodically in the Jordan River Watershed since 1919 at a site currently known as the Sunro Copper Mine (Burke 2012b). The mine is located underground in the vicinity of the old BC Hydro Forebay (Figure 2). Present access is by foot or ATV along a steep switchback trail on the west side of the old Forebay. Mine portals are visible along this trail though their entrances have been blocked with timbers and/or concrete. Previous access was by a 2.4 km tunnel initiating on the east side of the Jordan River opposite the current BC Hydro powerhouse. This access suffered cave-ins in 1977 and was subsequently blocked.

The site was first mined by Cominco in 1919 and 1920 (Burke 2012b). From 1929 to 1976, Cominco retained ownership but optioned mining operations to various other companies. In 1956 the 2.6 km access tunnel was dug and in 1961 a various improvements were made to the mining operation, including installation of a 15 cm tailings pipe which discharged into the ocean about 1 km east of the Jordan River mouth. The mine was then run at full production from 1962–1968 and from 1972–1974, when the price of copper declined. After this, mining continued at a lower rate until the access tunnel cave-ins in January 1977. The BC Ministry of Mines subsequently directed Cominco to close off the tunnel, which was achieved by installing a 34 m long rock plug at the point where the tunnel joins the mill room. The mine has remained inactive since that time though the property and mineral rights are currently up for sale with the potential to restart the mine (http://bcgold.com/mining-properties-for-sale/vancouver-island-121).

During the periods of full production, the Sunro Copper Mine produced an average of 1.9 metric tons of copper per year and smaller amounts of gold and silver (Burke 2012b). Copper concentrate from the mill room was transported by narrow gauge rail cars through the access tunnel to a storage

area at the portal. From here the containers were transported by truck to a loading dock in Cowichan Bay. An estimated 1.3 million metric tons of tailings and waste rock were generated during this period. Burke (2012b) reported that 95% of this material was deposited into the ocean just east of the Jordan River mouth. Tailings were pumped through a 15 cm diameter plastic pipe from the mill room to the access tunnel portal, and then an additional 1.5 km to a discharge point 12 m deep in the embayment east of the Jordan River outlet (see Figure 2).

Scientific documentation on the effects to local flora and fauna from the discharge of mine tailings into the ocean were not found, however, commentary from T'Sou-ke Nation elders mention that seafood and crabs were killed off for roughly 3 km on either side of the outfall (Recreation Resources Ltd 2001).

A noteworthy event for the Sunro Copper Mine was a major failure (possibly 1964), and the subject of a book by Loofbourow (1968). A large portion of the roof adjacent to the Forebay collapsed allowing the Jordan River to flow into the mine, flood the mill room, and travel down the access tunnel. Debris carried along with the water plugged the tunnel and the water burst out onto a ridge above the tunnel, washing out the portal and the trestle used to access the portal. Interestingly, commentary by elders in the Recreation Resources (2001) report discuss this event and how it resulted in the channel below the portal being filled with "4 feet" of gravel. The volume of sediment such that it prevented booming activities for 2 or 3 years during which logs had to be trucked to Sooke. They also mention that it was this wedge of sediment that was used to create the current dryland sort.

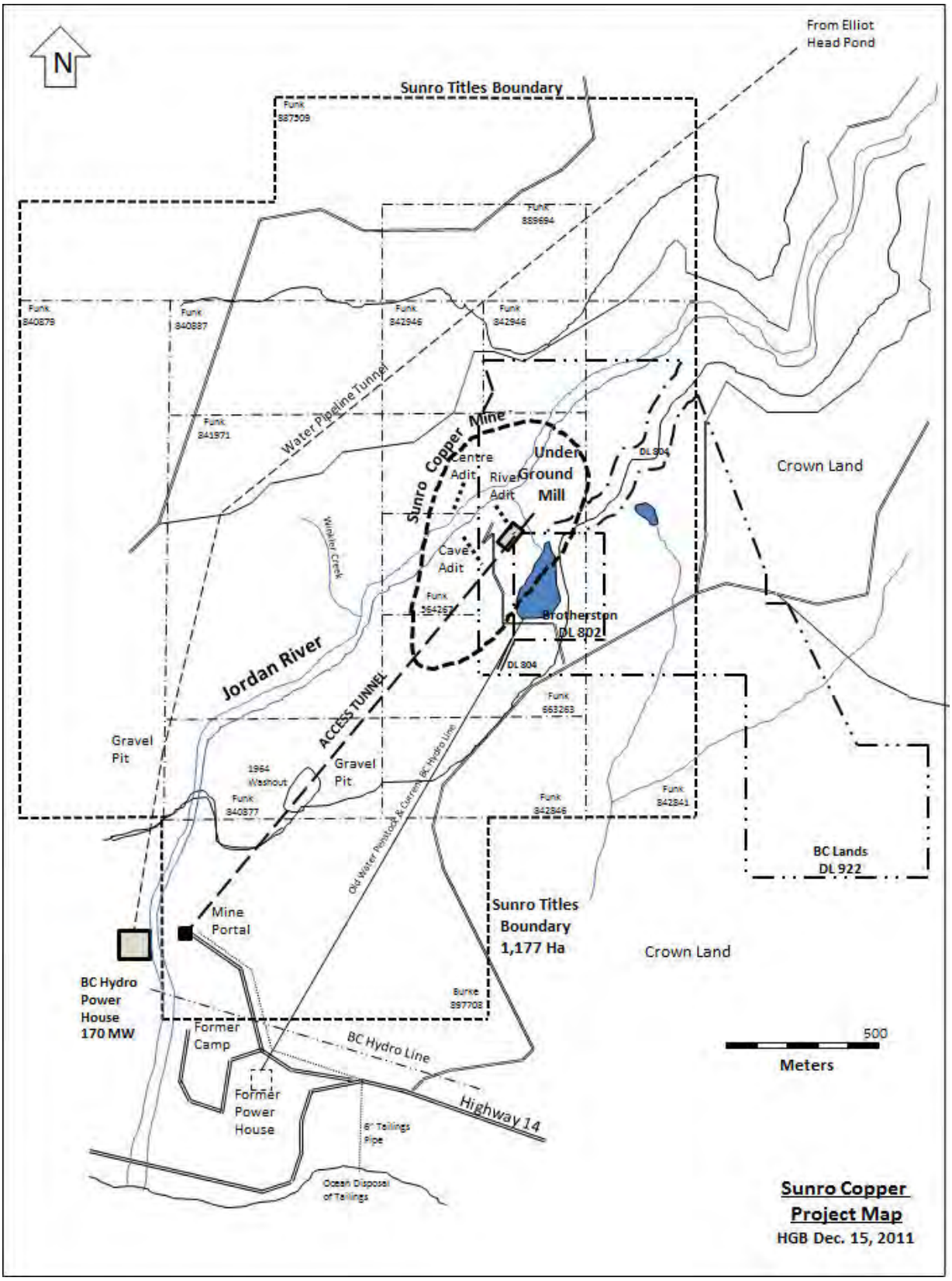

Figure 2. Location and features of the Sunro Copper Mine on the Jordan River (from http://www.sunrocopper.ca/location.htm**).**

3.5 Historical Presence of Anadromous Fish Species

Hirst (1991) reported that the anadromous reaches of the lower Jordan River once supported runs of coho, chum, and pink salmon while Griffith (1996) found evidence in Provincial records that steelhead and sea-run cutthroat trout were also once present. Fisheries and Oceans (DFO) escapement records for the Jordan River are irregular with many years reported as “no data” or “not inspected”. Nevertheless, years with data demonstrate that the lower reaches of the Jordan River once supported sizeable runs of coho, chum, and pink salmon (Figure 3). These early abundances are also consistent with levels of production described by T’Sou-ke Nation elders (Recreation Resources Ltd 2001).

Coho and chum salmon were present in the very first escapement records (1936) with annual spawner estimates of about 750 coho and 1500 chum (Figure 3). Coho numbers first declined in 1951, then dropped to just 25 fish from 1954 to 1965, followed by no fish observed from 1966 on. Chum salmon returns followed a similar pattern with an initial decline in 1951 and then no fish observed beginning in 1958. These findings suggest that some change or event occurred in the river or estuarine environments that compromised the ability of these species to successfully complete their life cycles. Assuming an average age of return of 3 years for coho salmon and 4 years for chum salmon, this change (or changes) may have occurred as early as 1947.

Specific information on the cause of these species declines was not present in available literature, though previous sections of this report offer several possible explanations for the decline of Jordan River coho and chum stocks:

- Sunro Mine activities— It is possible that the decline in salmon stocks was related to a change in mining activates or practices. In 1949 and 1950, Cominco optioned the property to Hedley-Mascot Mines, who installed electrical transformers and compressors near the old BC Hydro penstock intake (Burke 2012b). At this time access to the mine was by a 24 inch wide railway along the east side of the river and entry to the mine was via adits along the river bank. The company milled 600,000 tonnes. No information was found on how and where the tailings were disposed. Similarly, the access tunnel was built in 1956, but the tailings pipe was not installed until 1961, so there was another 5 years when tailings disposal was unknown. It is possible that disposal during these periods was along the river banks, either in the vicinity of the mine, or near the access tunnel entrance. It should be noted that the material observed along the river bank adjacent to the tunnel portal may be waste material from digging the tunnel as opposed to actual tailings (Burke 2012a).

  Also, though the timing is the 1960’s, the mine failure/washout event likely had an impact on fish stocks. Not only were large amounts of sediment deposited into the channel, but since the water washed through the mill room (where copper concentrate was produced) it may have conducted copper laden sediments to the lower river.

- Forestry activities— Anecdotal information suggests that the dryland sort was created in the mid 1960's. The salt marsh that was this location was reported to contain various tidal channels which were used by salmonids (Recreation Resources Ltd 2001). Loss of a major portion of the Jordan River estuary undoubtedly reduced the smolt production capabilities of the river, particularly for species such as coho and chum which rely heavily on this habitat prior to moving offshore. In addition, booming activities and dredging to maintain water access, would have further impacted production capabilities from the tidal reach.
- BC Hydro activities— At the time of the coho and chum declines, BC Hydro was still using the original generating station, tailrace, and manmade channel on the east side of the river, thus the 1971 rebuild does not appear to be linked with the decline of these two species. No flow data were available for this period of concern and thus it is unknown whether flow releases from Diversion Reservoir (or lack of) could have been a factor.

Though this may be in part a function of surveying the river too late, pink salmon do not appear in the escapement records until 1952 (Figure 3). The records show 8 years with pink returns (1952–1970), all on even years, with a mean escapement of roughly 3000 fish based on years with escapements. These fish were reported to spawn mainly in the 500 m long tailrace channel, though some spawned in the first 500 m of the mainstem in some years (Hirst 1991). Given the reliance of this species on the tailrace channel, and their subsequent disappearance in the year that BC Hydro moved operation to the west side of the river, it seems likely that decommissioning of the tailrace channel was a major factor for the loss of this run. In addition, following the rebuild, the Jordan facility began to be used as a peaking plant. This practice has resulted in radical fluctuations in flow below the new tailrace, potential changing from 0 to 67 m^3/s over short periods of time. This can cause stranding of adults and juveniles when flows suddenly drop, and downstream displacement of juveniles and disruption of adult spawning when flows suddenly increase. As well, the sudden and high energy of flows discharged from the turbine may have scoured and washed available spawning gravel out to the ocean. Clearly, the combination of moving the generating station to the mainstem, and operation of the facility as a peaking plant, both of which commenced in 1971, rendered the 900 m section of river downstream of the new tailrace a very inhospitable place for salmon production.

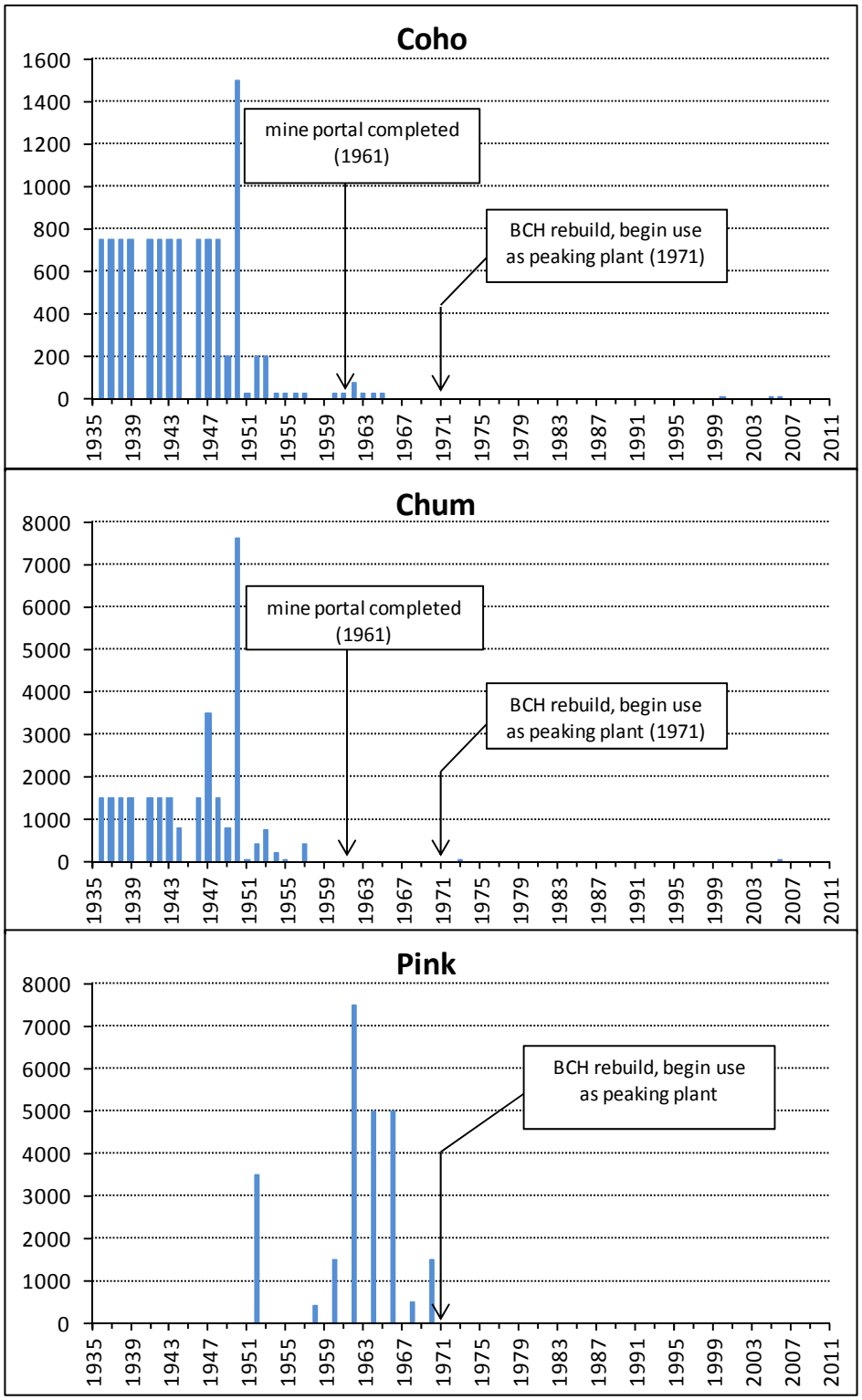

Figure 3. Escapement estimates for the Jordan River, 1936–2011. Sources: 1936–52 (Hirst 1991); 1953–2004 (nuSEDS database, DFO 2011); 2005–2008 JORMON#3 snorkel enumerations (Cascadia Biological Services 2006, 2007, 2009, 2010). Timing of changes in industrial operations also shown.

4. RESULTS

4.1 Description of Anadromous Reaches

The anadromous section of the lower Jordan River is comprised of three separate reaches, which when combined provide an accessible length of just under 1.6 km (based on digitizing in ArcView). The first reach spans the region of tidal influence, which includes from the mouth to half way up the first riffle immediately upstream of the Jordan Generating Station tailrace (Figure 4). This reach has a length of 920 m and is comprised of low gradient riffle-pool sequences at low tide. This reach has experienced extreme changes in flow over short periods (days) since 1971 when the generating station and tailrace were moved to the west side of the river and the Jordan facility commenced operation as a peaking plant. The sudden pulses of high flows have likely altered the channel morphology, banks, and substrate composition of this reach.

The second reach is 340 m long, consists of riffle-pool sequences, and terminates at the first major boulder cascades (Figure 4). This cascades is about 3 m in height and is likely a barrier to fish migration at low flows. Adult coho and steelhead can probably ascend this cascade at flows ≥ 0.8 m^3/s. In terms of substrate, cover, and gradient, this reach currently has the best spawning and rearing habitat within the anadromous portion of the river, however, until recently only the upper 20 m was inhabitable for rearing due to high concentrations of copper in the waters below this point. Cascadia Biological Services mapped available spawning habitat in this reach as part of the WUP monitor JORMON#3, and results indicated a total of 144 m^2 of habitat suitable for spawning (Cascadia Biological Services 2006, 2007, 2009). Coho fry from natural spawners were captured in the short section of unaffected water in the first year of the WUP JORMON#2 study (Burt 2006).

The third reach is 320 m in length and extends from the top of the 3 m cascade described above, upstream to a major boulder falls of about 4.6 m in height (Figure 4). This falls probably represents the upper anadromous limit and has been similarly judged by other fisheries surveys (Cascadia Biological Services 2001). This reach is comprised of a jumble of huge boulders (4 m diameter and greater) that appear to have originated from adjacent canyon walls. The river is very convoluted as it flows around, over, or under these huge boulders. Ascension within this reach by anadromous species would be difficult but may be possible under flows similar to those required for the partial barrier at the top of Reach 2. Rainbow trout were observed within the pools of this reach during the JORMON#2 study but no electrofishing was conducted to assess for potential presence of coho fry.

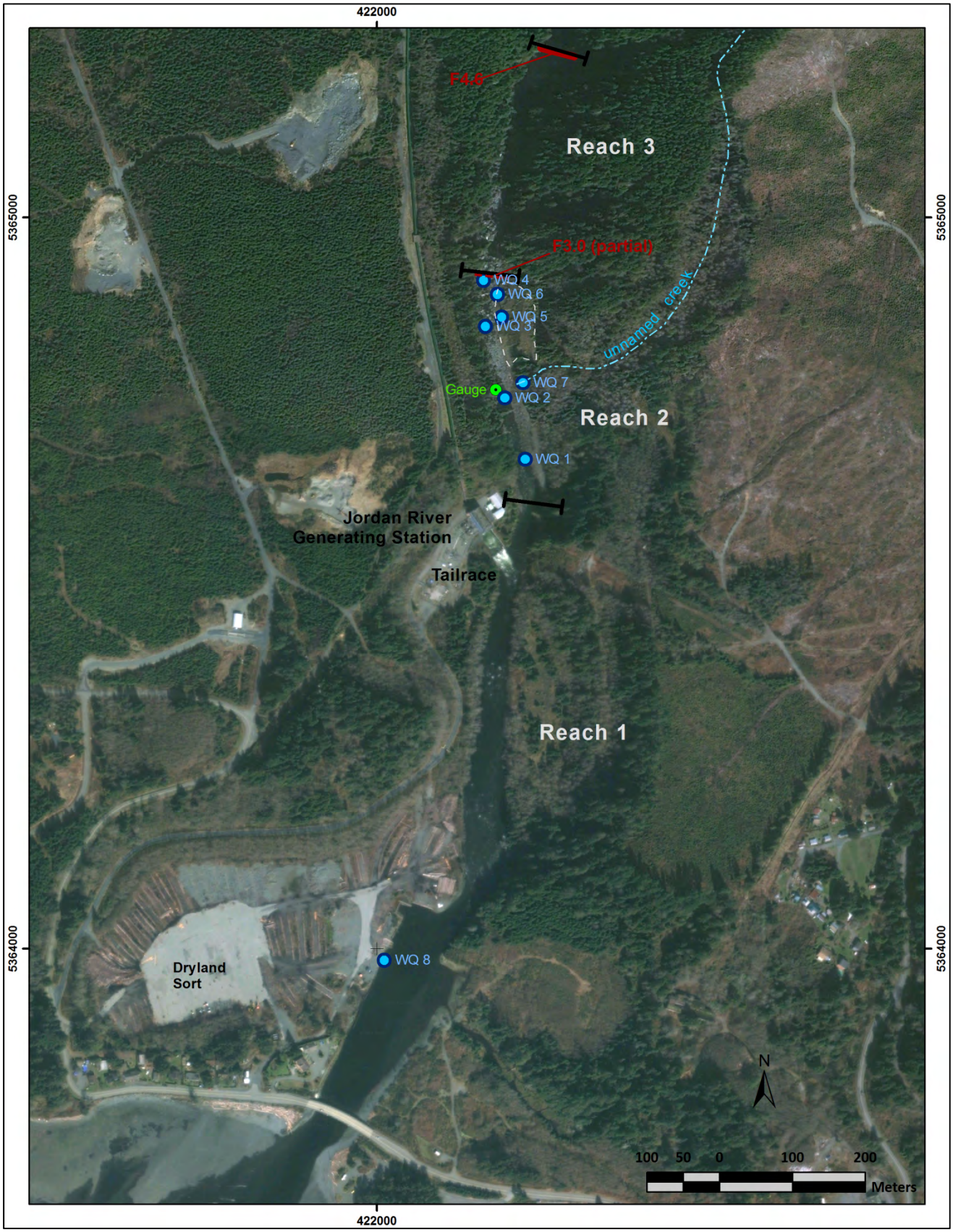

Figure 4. Map of Reaches 1, 2, and 3 of the Jordan River, the 8 water quality sites, and the flow gauging station.

4.2 Sources of Water Quality and Biological Data

The various sources that provided water quality and/or biological data pertinent to the anadromous reaches of the Jordan River are summarized in Table 1. The data span the years 1994 to 2012 encompassing periods before and after initiation of the fish flow release from Elliott Dam in January 2008. Generally these studies included regions of the Jordan Watershed beyond the anadromous reaches but the descriptions provided in the table relate only to anadromous portion.

Table 1. Sources of water quality and biological data pertaining to anadromous reaches of the Jordan river.

Year	Source	Pertinent WQ Data	Pertinent Biological Data
1994	(Griffith 1996)	3 site in anadromous section analyzed for general parameters (Alkalinity, pH, TDS, NFR, nitrates, phosphates). 2 sites were in copper affected zone, 1 was above. Also 2 sites received detailed analysis (32 elements including metals + hardness and pH). Of these, 1 was above and 1 was below the affected zone.	The study included assessment of juvenile salmonid abundance at 6 electrofishing sites in the anadromous section including 4 in the copper affected zone and 2 immediately upstream of this zone.
2000/01	(Lightly 2001)	Water samples were collected at the 3 incubation sites on Dec. 1, 2000 and analyzed for dissolved copper. A metal syringe was used to extract samples at depth for copper analysis however, cross-contamination rendered this data unusable.	6-7 incubation cassettes, each with 200 coho eggs were installed at each of 3 sites in Reach 2. 2 sites were in the copper affected zone, 1 was above.
2002/03	(Wright and Guimond 2003)	6 sites in Reach 2 were assessed for total extractable and dissolved copper. 3 sites were located in the copper affected zone, 1 was immediately upstream of this point, and 2 were at seepage points on the left bank. Water samples were collected periodically during Nov. 2002 to Feb. 2003.	6 incubation cassettes, each with 200 pink eggs, were planted at 3 sites. All 3 sites were in the copper affected zone. In addition, fry from Site 3 were put through a sea water challenge to assess whether exposure to copper produced developmental effects. Lastly, a controlled bioassay was conducted exposing pink fry to copper concentrations ranging from 0.026 mg/L to 0.166 mg/L.
2005–2010	(Burt 2006, 2007, 2008, 2009, 2010, 2011 (draft)) (JORMON#2)	2 sites in Reach 2 were assess for dissolved and total copper. 1 sites was in the copper affected zone and was above. Water samples were collected once per year (usually in September) for each of the 6 study years.	The study assessed juvenile salmonid abundance via electrofishing and included 3 sites in Reach 2. 2 sites were in the affected zone and 1 was immediately above this point.
2005–2010	(Cascadia Biological Services 2006, 2007, 2009, 2010, 2010 draft, 2011 draft) (JORMON#3)		The study included an annual incubation assessment at 5 sites with 3 in the affected zone and 2 above. 3 Incubators were installed at each site and each was filled with 200 eyed coho eggs. The study also included snorkel surveys in fall and spring periods to assess the abundance, distribution, and spawning success of coho and steelhead.

Year	Source	Pertinent WQ Data	Pertinent Biological Data
2012	Sunro Copper (Helmut Burke, pers. comm.)	On July 15, 2012 water samples were collected at 2 sites in the vicinity of the old mine (adjacent to the old Forebay), at 1 site in the affected zone of Reach 2, and at 1 site in Reach 1 (by the dryland sort). Samples were analyzed for total copper.	
2012	BC hydro sampling (Ian Dodd, Allister McLean)	On September 24 2012, 2 replicate water samples were collected at 2 sites in the copper zone, 1 site immediately upstream, and 1 in the unnamed tributary. Samples analyzed for dissolved and total copper.	

In the process of compiling the data from the above sources, it was found that many of the studies selected the same geographic locations for their monitoring sites but sometimes used different names. To avoid confusion I mapped the various water quality and biological monitoring sites and named them using an abbreviation of the monitoring category followed by a number. For example, water quality monitoring sites were named “WQ x”, incubation sites “Inc x”, and electrofishing sites “EF x”, where x represents a sequential number, usually increasing with upstream progression.

4.3 Summary of Copper Findings

Copper was assessed at 8 sites in the anadromous reaches (Figure 4) with some sites receiving multiple readings over a number of months (Wright and Guimond 2003) or over a period of years (Burt 2011 (draft)). Depending on the study, copper was reported as dissolved copper, extractable copper, total copper, or a combination of these. The advantages and disadvantages of these three forms of copper are discussed in the Provincial Water Quality Criteria Overview Report (Ministry of Environment 1998). In general, dissolved copper readings are less than extractable values, which are less than total copper values. In the case of the lower Jordan River, there is not much difference between the three forms of copper, which the Criteria Report indicates is common in oligotrophic waters. This facilitates comparison among studies when different forms of copper were used.

Table 2 presents a summary of available copper data organized in chronological order and segregated according to sites located adjacent to and downstream of the “tailings” bank, versus sites located upstream of this bank. Where possible, discharge at the time of sampling was assigned to each reading based on results from the Inflow Monitoring Program (JORMON#1, Burt and Hudson 2012). These data show that levels of copper in the mainstem adjacent and downstream of the “tailings” bank (copper affected zone) ranged from 0.007 to 0.127 mg/L over the period of record (all forms combined). The large fluctuation in readings within this zone appears to be inversely related to river flow at the time the samples were taken. In comparison, values from upstream of the “tailings”

ranged from <0.001 to 0.005 mg/L and flow had no apparent affect on copper concentration. Site WQ 4, which is immediately above the tailings, and which studies tended to use as a control, had slightly elevated copper concentrations compared with water quality sites further up the river indicating that this location had small amounts of copper input.

Table 2. Summary of available copper data organized by sites adjacent to and downstream of the affected zone and upstream of this zone.

Source	Date	Original Site Name	Common Site Name	M1 Q (m^3/s)	Dissolved Cu (mg/L)	Extractable Cu (mg/L)	Total Cu (mg/L)	Comments
Sites adjacent and downstream of the "tailings" bank								
Griffith (1996)	19/08/1994	#2	WQ 1	0.024			0.127	
Lightly (2001)	02/12/2000	#3	WQ 1	3.000	0.014			
	02/12/2000	#2	WQ 3	3.000	0.012			
Wright and Guimond (2003)	01/11/2002	#1	WQ 1	0.027	0.069	0.070		
(surface samples only)	01/11/2002	#2	WQ 2	0.027	0.064	0.064		
	01/11/2002	#3	WQ 3	0.027	0.063	0.065		
	13/11/2002	#1	WQ 1	0.661	0.012	0.013		
	13/11/2002	#2	WQ 2	0.661	0.015	0.017		
	13/11/2002	#3	WQ 3	0.661	0.011	0.012		
	05/12/2002	#1	WQ 1	0.130		0.079		
	05/12/2002	#2	WQ 2	0.130		0.084		
	05/12/2002	#3	WQ 3	0.130		0.056		
	05/12/2002	#5 (tailings)	WQ 5	seepage		0.554		
	31/01/2003	#1	WQ 1	3.052		0.010		
	06/01/2003	#2	WQ 2	0.627		0.024		
	06/01/2003	#3	WQ 3	0.627		0.017		
	06/01/2003	#6 (mine pipe)	WQ 6	seepage		0.445		
Burt (2011 (draft))	27/09/2005	EF02	WQ 1	0.016	0.060		0.064	
(JOR MON#2)	01/09/2006	EF02	WQ 1	0.012	0.086		0.086	
(average of 2-3 replicates)	09/09/2007	EF02	WQ 1	0.035	0.040		0.049	
	28/08/2008	EF02	WQ 1	2.948	0.007		0.008	Begin fish flow release
	23/09/2009	EF02	WQ 1	0.485	0.015		0.016	(generally 0.3-0.4 m3/s)
	21/09/2010	EF02	WQ 1	1.059	0.009		0.009	
Sunro Copper sampling	15/07/2012	at log sort	WQ 7				0.009	
	15/07/2012	d/s of tunnel	WQ 3				0.012	
BC Hydro sampling	24/09/2012		WQ 1	0.340	0.012		0.015	
	24/09/2012		WQ 3	0.340	0.009		0.013	
	24/09/2012	Unnamed ck.	WQ 7	creek	0.012		0.024	
Minimum				**0.012**	**0.007**	**0.010**	**0.008**	
Maximum (river samples)				**3.052**	**0.086**	**0.084**	**0.127**	**Excludes seepage samples**

Source	Date	Original Site Name	Common Site Name	M1 Q (m^3/s)	Dissolved Cu (mg/L)	Extractable Cu (mg/L)	Total Cu (mg/L)	Comments
Sites upstream of "tailings" bank								
Griffith (1996)	19/08/1994	#3	WQ 4	0.024			0.003	
Lightly (2001)	02/12/2000	#1	WQ 4	3.000	<0.005			
Wright and Guimond (2003)	01/11/2002	#4	WQ 4	0.027	0.003	0.003		Surface samples only
	13/11/2002	#4	WQ 4	0.661	0.004	0.004		
	14/02/2003	#4	WQ 4	0.037		0.003		
Burt (2011 (draft))	27/09/2005	EF04	WQ 4	0.016	<0.001		0.002	
(JOR MON#2)	27/09/2005	EF06	WQ 11	0.016	<0.001		<0.001	
(average of 2-3 replicates)	27/09/2005	EF14A	WQ 13	0.016	0.001		0.001	
	01/09/2006	EF04	WQ 4	0.012	0.001		0.002	
	01/09/2006	EF06	WQ 11	0.012	0.001		0.001	
	01/09/2006	EF13	WQ 12	0.012	0.001		0.001	
	01/09/2006	EF14A	WQ 13	0.012	0.001		0.001	
	09/09/2007	EF04	WQ 4	0.035	0.002		0.004	
	09/09/2007	EF06	WQ 11	0.035	0.001		0.001	
	09/09/2007	EF13	WQ 12	0.035	0.001		0.001	
	09/09/2007	EF14A	WQ 13	0.035	0.001		0.001	
	28/08/2008	EF04	WQ 4	2.948	0.004		0.005	Begin fish flow release
	28/08/2008	EF06	WQ 11	2.948	0.002		0.002	(generally 0.3-0.4 m3/s)
	28/08/2008	EF13	WQ 12	2.948	0.002		0.002	
	28/08/2008	EF14A	WQ 13	2.948	0.003		0.002	
	23/09/2009	EF04	WQ 4	0.485	0.003		0.003	
	23/09/2009	EF06	WQ 11	0.485	0.002		0.001	
	23/09/2009	EF13	WQ 12	0.485	0.001		0.001	
	23/09/2009	EF14A	WQ 13	0.485	0.001		0.001	
	21/09/2010	EF04	WQ 4	1.059	0.003		0.003	
	21/09/2010	EF06	WQ 11	1.059	0.001		0.001	
	21/09/2010	EF13	WQ 12	1.059	0.001		0.001	
	21/09/2010	EF14A	WQ 13	1.059	0.001		0.001	
Sunro Copper sampling	15/07/2012	d/s of mine	WQ 10				0.001	
	15/07/2012	u/s of mine	WQ 11				0.002	
BC Hydro sampling	24/09/2012		WQ 4	0.340	0.001		0.002	
Minimum				**0.012**	**<0.001**	**0.003**	**<0.001**	
Maximum				**3.000**	**0.004**	**0.004**	**0.005**	

Geographic Variations in Copper

The copper monitoring completed as part of the Jordan River Fish Index Study (JORMON#2) provided the best dataset on variations in copper at different locations on the mainstem between the generating station and Elliott Dam. Water samples were collected annually in late August or

September from 2005 to 2010 and were analyzed for both dissolved and total copper (Figure 5). Results for 2005–2007 show copper levels before initiation of the fish flow release from Elliott Dam while 2008–2010 show levels after the flow release. WQ 1 is located in the copper affected zone while WQ 4 is immediately above this zone. The remaining sites are located in the mainstem adjacent to the Forebay (WQ 11) to roughly 800 m downstream of Elliott Dam (WQ 13). These data show that dilution from the fish flow releases reduced copper levels in the affected zone from 0.040–0.085 mg/L before the release to 0.007–0.016 mg/L after the release. Upstream sites (WQ 11 to WQ 13) probably represent natural copper levels in the river and these ranged from <0.001 to 0.003 mg/L with 6-year means of 0.001 mg/L for both dissolved and total copper.

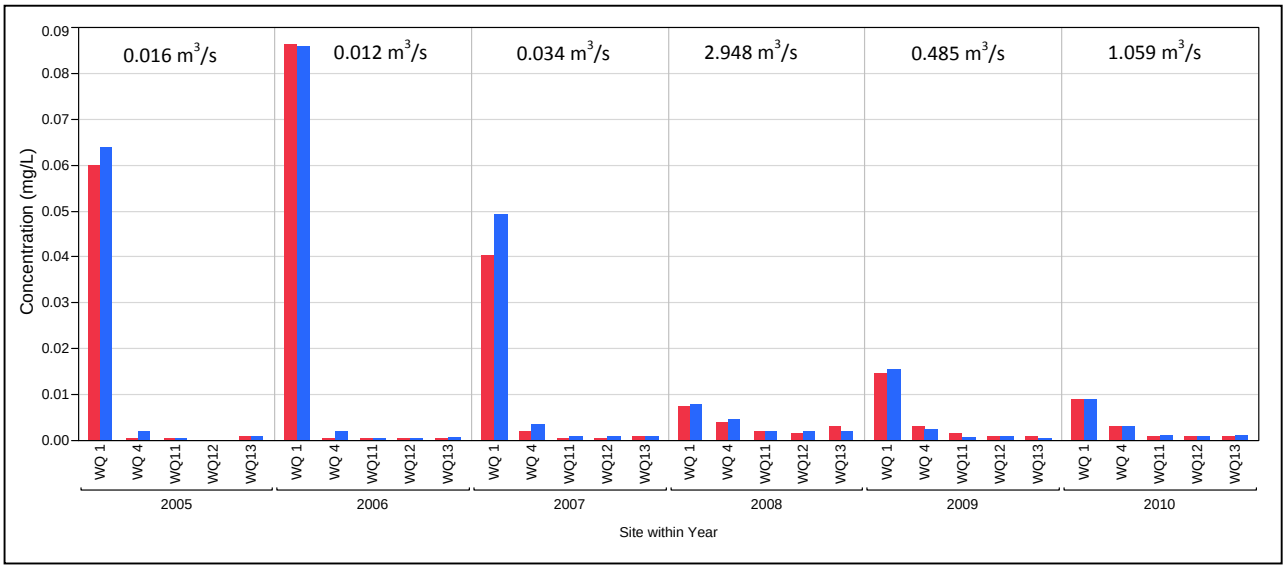

Figure 5. Dissolved and total copper by site from the Lower Jordan River Fish Index Study (JORMON#2). Also shown are flows at the lower river gauging station at the time of sampling. Note: WQ11 to WQ13 are located from adjacent the Forebay upstream to just below Elliott Dam. Source: Burt (2011 (draft)).

Influence of Flow on Copper Concentration

When copper readings from the various studies were paired with stream discharge, it became apparent that there was some form of inverse relationship between discharge and copper concentration. This is as expected since the copper is believed to originate from a point source (a specific section of bank), and greater flows in the river should result in greater dilution of the copper inputs. The nature of this relationship was explored by regressing readings of dissolved copper on stream discharge. Stream discharge was for the lower river gauging station and was obtained from the Jordan Inflow Monitoring Program database (Burt and Hudson 2012) which spanned the years 2005 to 2010. In addition, discharge for post 2010 readings was calculated from the stage/discharge

equation as long as a staff gauge reading was taken at the time of the copper sampling. For copper records before 2005, modelled estimates of flow were used. This relied on analyses by Burt and Hudson (2012) where BC Hydro's total system inflows to Elliott Headpond (TSI's), watershed areas, and a modelled/measured correction factor were used to estimated flows at the lower river gauging station. It should be noted that the modelled flows were based on mean daily values, and so when copper samples were taken on rising or falling freshets there is potentially a degree of error in the assigned flow value.

Another aspect of this analysis is that data prior to the fall of 2006 were regressed separately from data beginning in summer 2007 (i.e., individual curves were fitted for each group). This was done due to a series of extreme freshet events that occurred during winter of 2006/07 which resulted in heavy scour in the lower river[1]. Much of the stream bed was mobilized during this event and any banks with loose material were truncated to a height of about 3 m. The following summer (2007), copper levels in the affected zone appeared to be less than previous years and it was speculated that some of the source material may have been washed out to the ocean (Burt 2010).

The results of the regression analysis are shown in Figure 6. The best fit is for both data sets was a power curve. The two equations have similar exponents but differ in the first parameter. The post flood curve predicts a lower dissolved copper value for a given discharge than the pre flood curve. Examples of predicted values relevant to operation of Elliott Dam fish flow releases include the following:

	Pre Flood Predicted Cu (mg/L dissolved Cu)	Post Flood Predicted Cu (mg/L dissolved Cu)
0.25 m^3/s:	0.029	0.017
0.30 m^3/s:	0.027	0.016
0.40 m^3/s:	0.024	0.014

The first example is the prescribed flow release in the Jordan WUP (0.25 m^3/s) while the second and third examples are the typical range of actual flow releases due to the control valve being locked in the full open position. Under these flows, levels of dissolved copper in the affected zone are currently predicted to range between 0.015 and 0.017 mg/L during summer base flows.

[1] Three major spill events occurred at Elliott Dam. Peak spills for each event were 459 m^3/s on 6 November 2006, 564 m^3/s on 2 January 2007, and 636 m^3/s on 11 March 2007.

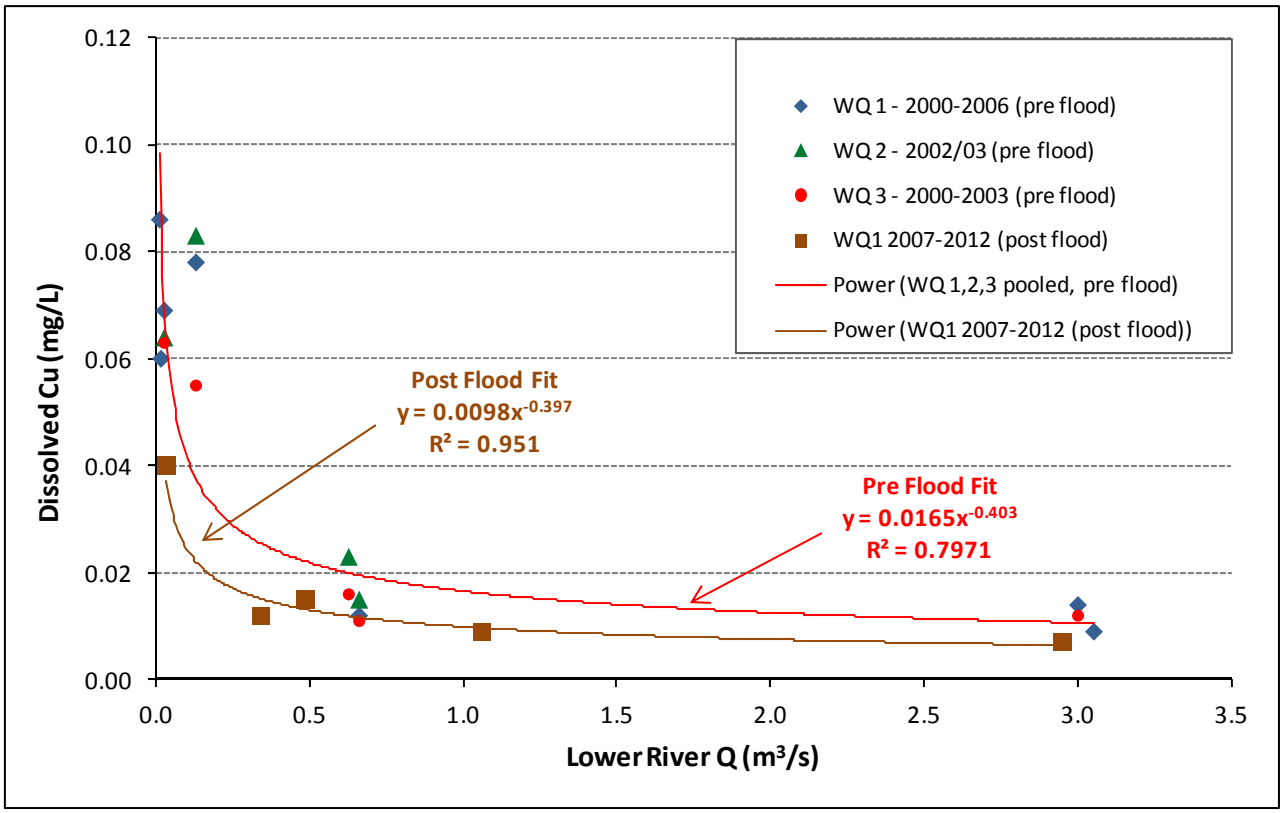

Figure 6. Relation between dissolved copper in the affected zone and stream discharge at the lower river gauging station. Two curves were generated: one for data prior to the winter 2006/07 floods, and one for data after this event. Both are power curves and were generated in Excel.

Possible Sources of Copper

Previous studies on the lower Jordan River have often assumed that the 80 m long section of exposed bank on the east side of the river at the top of reach 2 is comprised of tailings from historic copper smelting operations, and that this is the source of copper contamination in the lower river. However, in March 2012 Burke (2012a) examined this site and noted that the material was darker and courser than tailings and concluded that the bank deposit was more likely comprised of excavated material from when the access tunnel was dug in 1956. Burke estimated the dimensions of the deposit to be 80 m long x 16 m wide x 7 m deep, indicating a total volume of 9000 m^3, which he felt was consistent with the volume of material removed to create the access tunnel. If this is true, the source of copper contamination may be from other locations nearby, and the bank deposit may simply be a conduit for seepage from these other sources. A photo of this bank deposit is provided in Figure 7.

Figure 7. Photo of the mine deposit on the left bank at the top of Reach 2. Burke (2012a) estimated the dimensions of this deposit to be 80 m long x 16 m wide x 7 m deep. This photo was taken by the author on Sept. 21, 2005. In the winter of 2006/07 extreme high flows cut away the base of this bank up to the foot of the first trees in the background.

Another possible source of copper is seepage from the mine access tunnel. At the far end of this tunnel is a large cavern where the ore was milled and processed to produce copper concentrate. The tunnel was plugged in 1977 and the cavern is currently backfilled with water to a depth of about 3 m. Given the purpose of the mill room, it is possible that the water in the cavern is high in copper, and may be seeping down the tunnel to ultimately end up in the river adjacent to the portal.

From this discussion it is apparent that there is uncertainty as to the source of the copper contamination in the lower river. From Wright and Guimond (2003), we do know that there are at least two seepages coming out the bank deposit, and that these have high levels of copper (0.554 mg/L at WQ 5 and 0.445 mg/L at WQ 6, Table 2). What we do not know, is whether the high copper levels in these seepages are due to the bank material itself, or originate from points beyond the bank (such as the mine portal), or perhaps even derived from both locations. This is a significant deficiency in our understanding since a solution to the copper contamination in the river cannot be investigated until this is known.

4.4 Summary of Biological Findings

A synopsis of available biological studies on the lower Jordan River was previously given in Table 1. Together, available studies encompass spawning, incubation, and rearing of salmonids in the lower Jordan River and provide insights on the success of these life stages before and after initiation of the fish flow release at Elliott Dam. The following is a summary of the performance of these life stages within and upstream of the copper affected zone of the lower Jordan River.

Spawning

Information on adult salmonid use in the anadromous reaches of the lower Jordan River is extremely limited as DFO ceased their annual escapement counts to the river around 1979 (DFO 2011). The exception was 6 surveys conducted between October 18 and November 2, 2000, when a peak count of 8 coho was recorded for the lower river. Anecdotal information is also provided by Jordan River Generating Station staff who annually observe small numbers of pink, coho, chum, and sockeye salmon in the generating station tailrace (Dwayne Walsh, pers. comm.). These fish tend to be attracted into the tailrace during generation, and subsequently become trapped until generation commences again.

From 2005 to 2011, Cascadia Biological Services conducted annual snorkel enumerations in Reach 2 of the Jordan River as part of the Jordan River WUP monitoring (JORMON#3, Cascadia Biological Services 2011 draft). During the 6 years of the program, 5 surveys were conducted annually from September 21 to December 21, targeting the coho salmon spawning period, and another 3 surveys were conducted from April 15 to May 31, targeting the steelhead spawning period. It should be noted that these surveys were too late to enumerate pink salmon spawning. Another limitation is that the surveys did not include Reach 1, which was historically used by pink and chum salmon, and which has received some spawning activity in fall 2012 (Dwayne Walsh, pers. comm.).

The results of JORMON#2 are shown for the fall salmon surveys in Chart A of Figure 8, and for spring steelhead surveys in Chart B of the same figure.

Overall, counts from both the fall and spring surveys were very low, ranging from 0 – 4 fish for coho salmon, 0 – 13 fish for chum salmon, and 0 – 2 fish for steelhead trout. The results do not indicate any change in abundance following the flow release in 2008, though given age of return for these species, potential benefits might not be expected until at least fall 2011 returns.

Cascadia felt that spawner abundance was limited by a shortage of gravel (144 m^2 in Reach 2), and that freshwater survival was hampered by scour of redds by high flows during spill events. Also with such small numbers of adults returning to the river, Cascadia hypothesized that loss of individuals by predation (seals, eagles, and otters) may have a significant impact on spawning success. This is somewhat aggravated by entrapment of fish in the tailrace, which provides an easy meal for local otters. The study did not provide any evidence that spawning was adversely affected by ambient levels of copper in Reach 2.

Despite the various factors weighing against freshwater survival, there are years when winter flows are unlikely to scour redds. For example, excluding the extreme spills that occurred during the 2006/07 winter, Elliott Dam flow records indicate that between 2005 and 2011, there were 2 winters with no significant spills (2005/06, and 2009/10), and 3 winters with moderate spills (2007/08, 2008/09, and 2010/11, peak flow 133–174 m^3/s). Successful spawning was also demonstrated by the juvenile Fish Index Study (JORMON#2) which captured coho fry immediately upstream of the copper affected zone in September 2005 (Burt 2006). These fish had to be progeny from natural spawning as the WUP incubation monitoring program (which used coho eggs) did not commence until the fall of 2005.

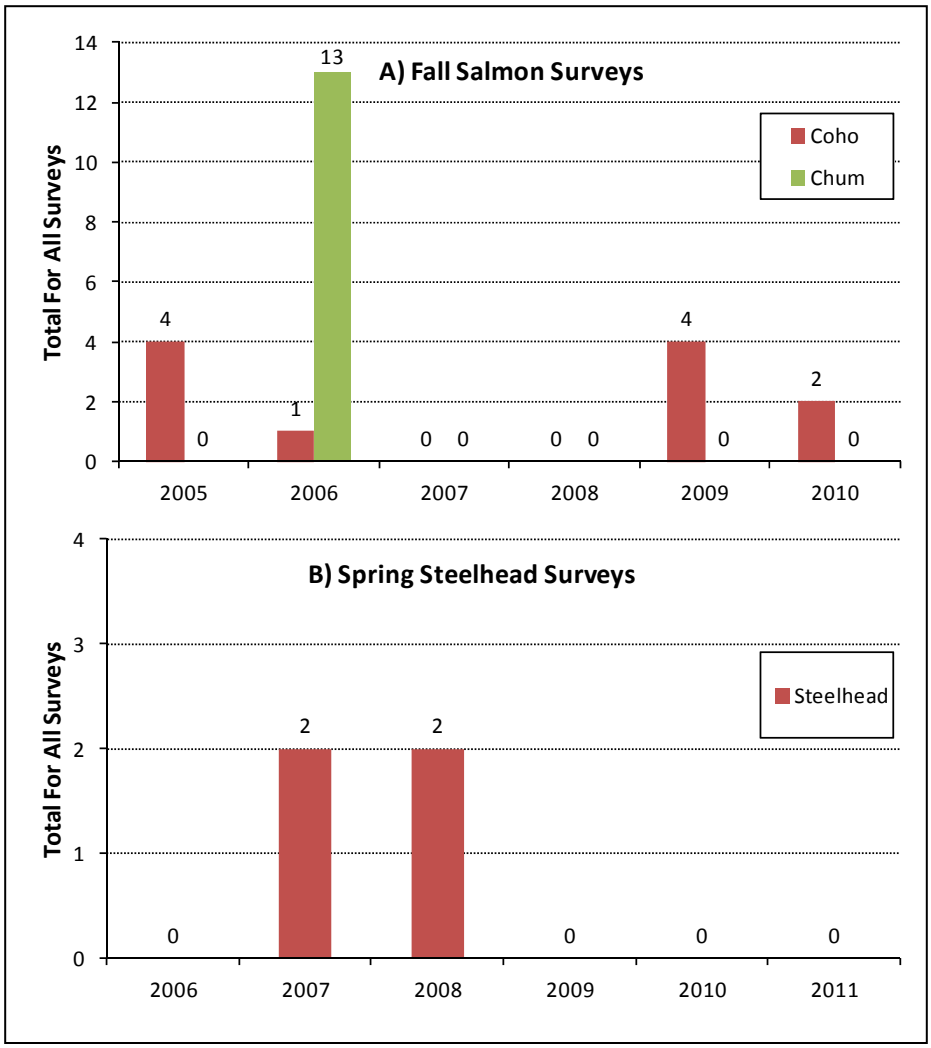

Figure 8. Summary of fall (Chart A) and spring (Chart B) snorkel survey enumerations conducted on Reach 2 of the Jordan River from 2005 to 2011. Adapted from JORMON#3 (Cascadia Biological Services 2011 draft)

Incubation

Information on incubation success are available from 3 studies spanning the winters of 2000/01 to 2010/11. All together, there were 6 sites where incubators were planted, the locations of which are shown in Figure 9. A synopsis of the study design of these investigations is provided in Table 3. All 3 studies used Jordan-Scotty Incubators though JORMON#3 used the model with fry escape holes whereas previous investigations used models without this feature. These incubators house 200 eggs with each egg in its own pocket equipped with an inlet and outlet pore for water exchange. The study by Wright and Guimond (2003) used pink eggs while the other 2 studies used coho eggs.

Figure 9. Map of Reach 2 showing the mine deposit, locations of the 6 incubation sites and of the 5 electrofishing sites.

Table 3. Study design of the 3 incubation investigations conducted in the anadromous reaches of the Jordan River.

Study Period	Source	Study Design
2000/01	Lightly (2001)	• Jordan-Scotty incubators (JSI's) without fry escape holes. • Incubation sites 2, 3 and 5 in Figure 9. • 6 - 7 incubators per site. • 200 green coho eggs per incubator • Replanted after each check
2002/03	Wright and Guimond (2003)	• JSI's without fry escape holes. • Incubation sites 2, 3 and 4 in Figure 9. • 6 incubators per site. • 200 eyed pink eggs per incubator • Replanted after each check
6 years from 2005/06 to 2010/11	JORMON#3 (Cascadia Biological Services 2006, 2007, 2009, 2010, 2010 draft, 2011 draft)	• JSI's with fry escape holes. • Incubation sites 1, 2, 3, 5, and 6 in Figure 9. • 3 incubators per site. • 200 eyed coho eggs • Not replanted after each check

Results from these 3 studies are summarized in Figure 10. With the exception of Sites 2 and 3 in 2002/03, egg-to-emergence survival rates were high, generally > 90%. The results also show no significant difference between sites within the copper zone and sites upstream of this zone. Lastly, there is no change in incubation survival following initiation of the flow release in January 2008. Thus, these data suggest that levels of copper in the lower Jordan River during winter/spring periods are not impacting survival during the egg-to-emergence stages of development. This is consistent with the conclusions drawn by the authors of these 3 studies.

For the cases in Figure 10 with low survival, the authors felt that results were due to siltation on or within the cassettes, and/or low intergravel flow/dissolved oxygen conditions. In particular both the 2002/03 and JORMON#3 studies noted siltation at Site 3, which was believed to originate from the unnamed tributary on river left. The JORMON#3 study moved this site 15 m upstream in Year 2 to avoid this issue.

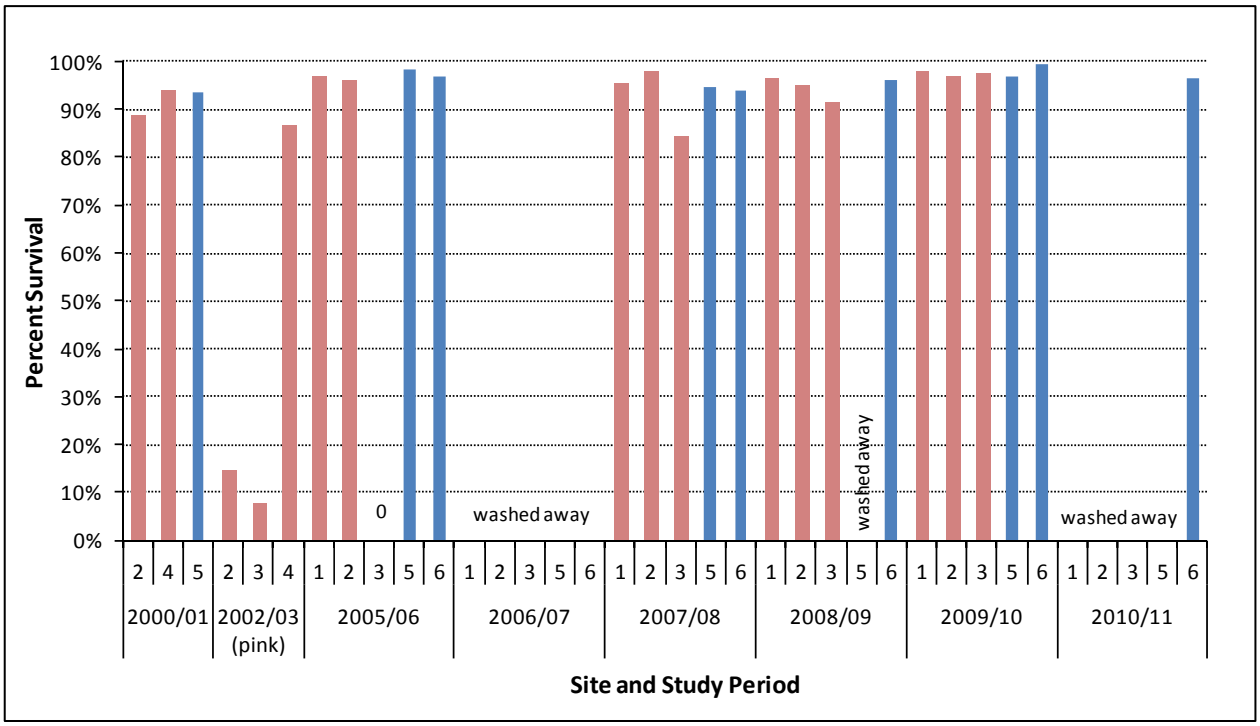

Figure 10. Egg-to-emergence survival from incubation studies conducted within anadromous reaches of the Jordan River. Tan bars represent survival at sites in the copper affected zone while blue bars show survival at sites upstream of this zone. Coho salmon eggs were used with the exception of 2002/03, when pink salmon eggs were used. Sources: 2000/01 from Lightly (2001), 2002/03 from Wright and Guimond (2003), 2005/06 to 2010/11 from Cascadia (2011 draft).

Freshwater Rearing

Studies that provide information on freshwater rearing within the anadromous reaches of the Jordan River include the juvenile abundance inventory by Griffith in 1994 (Griffith 1996), and similar inventories conducted by the author from 2005 to 2010 (JORMON#2 Fish Index Study, Burt 2011 (draft)). In addition, a seawater challenge and copper toxicity bioassay was conducted on pink fry as part of the incubation study by Wright and Guimond (2003).

The inventory by Griffith (1996), and the 6-year JORMON#2 study (Burt 2011 (draft)) employed total removal electrofishing techniques using 2 or more passes to estimate the population of fish within netted enclosures. A portion of the JORMON#2 sites deviated from this approach and used 1-pass efforts with the contained population estimated from a regression relationship between pass 1 catches and total population estimates from multi-pass sites. Griffith's inventory included 2 sites in the copper affected zone (EF01 and EF02) and 2 above this zone (EF04 and EF05). The JORMON#2 study included 2 or 3 sites in the copper zone (EF01, EF02 and EF03) and 1 site immediately upstream of this zone (EF04). Locations of the 5 sites are shown in Figure 9. Site EF01 was in the zone of tidal influence and was 150 m downstream of the tailrace in Griffith's study, and 75 m upstream of the tailrace in JORMON#2.

Results of these inventories, expressed in terms of fish density (fish/100 m^2), are reported for rainbow trout in Figure 11. Chart A shows the abundances of age 0^+ rainbow, Chart B of age 1^+ rainbow, and chart C of ages 2 and 3^+ rainbow. Densities at sites in the copper affected zone were given tan bars to distinguish them for densities at sites upstream of this zone (blue bars).

In Griffith's 1994 inventory, and in the first 2 years of the Fish Index Study (2005 and 2006), no fish were found at sites within the copper affected zone (Figure 11). When the author encountered this in 2005, the entire length of Reach 2 was subsequently walked with the electrofisher turned on targeting prime rearing spots for the whole length. This effort encountered no fish until reaching the location of EF04 where various age classes of rainbow, as well as coho fry, were found. In 2007, rainbow fry (i.e., age 0^+) were encounter at sites within the copper zone (other age classes were absent). However, these fish were extremely slender suggesting that either food was scarce, or that development/weight gain was adversely affected by ambient copper concentrations. In 2008, and continuing in 2009 – 2010, rainbow trout were once again encountered at sites in the copper zone and had better girth than observed in 2007. In addition, other age classes of rainbow were found for the first time at sites in the copper zone (Charts B and C). Burt (2011 (draft)) attributed the 2007 results to removal of some of the copper source material by the extreme floods the previous winter, and the 2008 – 2010 results to this plus dilution of copper inputs by the fish flow releases which commenced in January 2008.

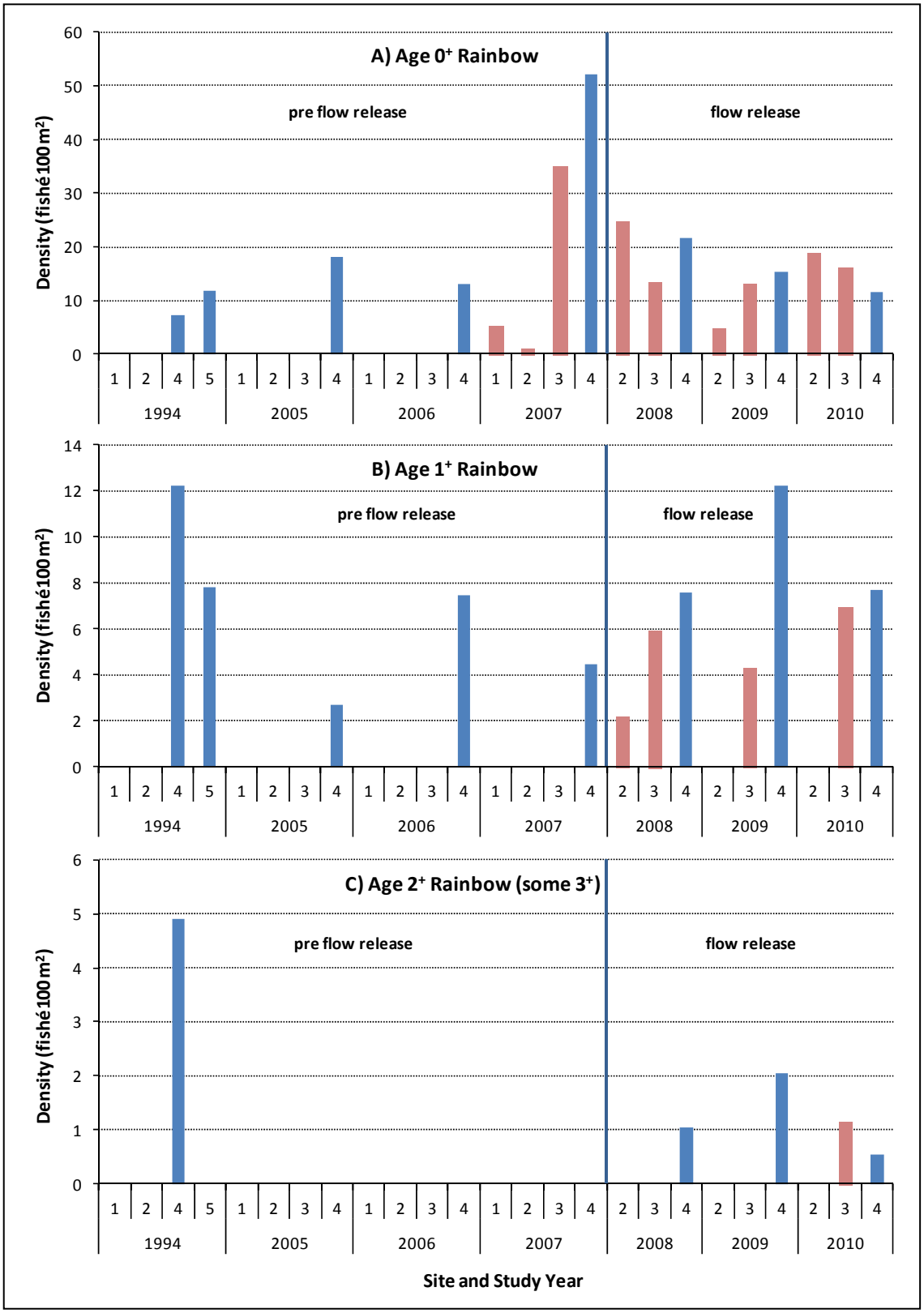

Figure 11. Density of rainbow trout (fish/100 m^2) at electrofishing sites in the anadromous reaches of the Jordan River. Each chart presents a different age class. Tan bars show abundances at sites in the copper affected zone; blue bars show abundances at sites upstream of this zone. Data adapted from Griffith (1996) and Burt (2011 (draft)).

In addition to rainbow, coho fry were captured in 2 of the 5 years prior to the flow release, all at EF04, located immediately upstream of the copper zone (Figure 12). Following initiation of the flow release, coho fry continued to be captured at this site in the 3 remaining years of JORMON#2, but were also captured within the copper zone for the first time in 2009 (Site EF03). Furthermore, these fry displayed similar girth to those at EF04 (condition factor was 1.19 compared with 1.17 at EF04), suggesting copper levels were low enough at EF03 to allow normal growth and development of coho fry. As to the source of these coho, the fry captured in 2005 had to be from natural spawning, however, those captured in subsequent years may have originated from the incubation study (JORMON#3) which ran concurrent to the Fish Index Study.

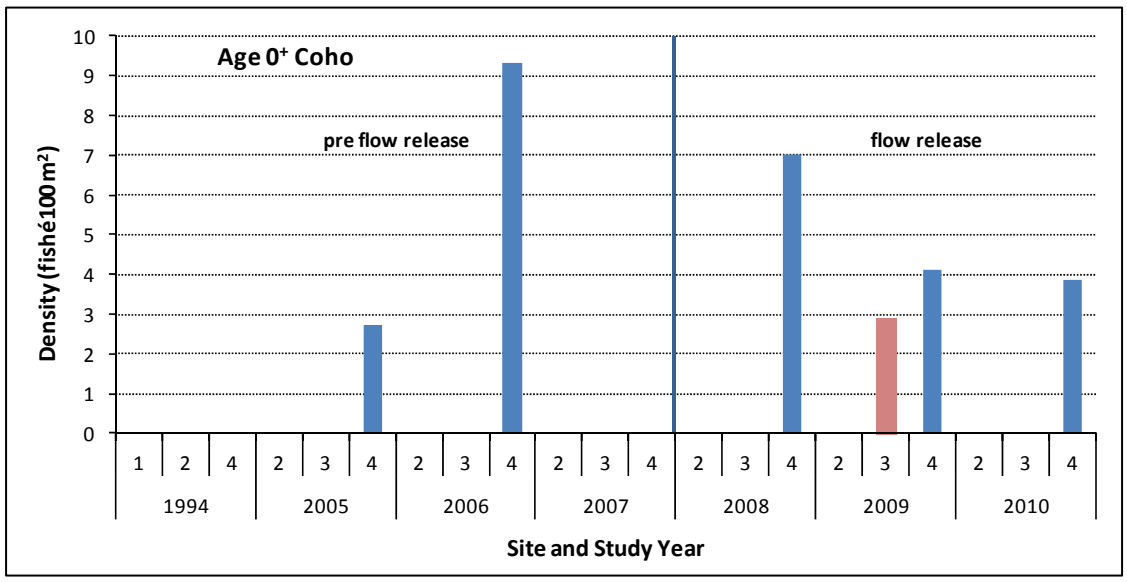

Figure 12. Density of coho fry (fish/100 m^2) at electrofishing sites in the anadromous reaches of the Jordan River. Tan bars show abundances at sites in the copper affected zone; blue bars show abundances at sites upstream of this zone. Data were adapted from Griffith (1996) and Burt (2011 (draft)).

From the above discussion, it is apparent that fish condition factor[2] offers an addition approach for assessing how well rearing fish are performing in waters with elevated levels of copper, particularly when compared with condition factors for fish upstream of the copper affected zone (control). This was done for 0^+ rainbow trout captured at Site 1 though 4 spanning the years 2005 to 2010 (data were extracted from the Fish Index Study database for this analysis). The results of this

[2] Fish "condition factor" is a measure of the relationship between a fishes weight and length, and as such, fish with a larger girth at a given length have a higher condition factor than thinner fish of the same length. The condition factor used in JORMON#2 was Fulton's Condition Factor, calculated as CF = Wt * 10^5/(Fork Length)^3. A typical Fulton's CF for BC rainbow trout is 1.05.

compilation are shown in Figure 13. For comparative purposes, the Provincial default condition factor is also shown (1.05). These data show that rainbow fry in the copper zone had poor condition factor in 2007 (site 3 was significant relative Site 4; lack of significance for Sites 1 and 2 was due to small sample size), whereas in 2008 to 2009 condition factors of fish in the copper zone were similar or even greater than Site 4 (Site 3 was significantly greater than Site 4 in 2009 and 2010).

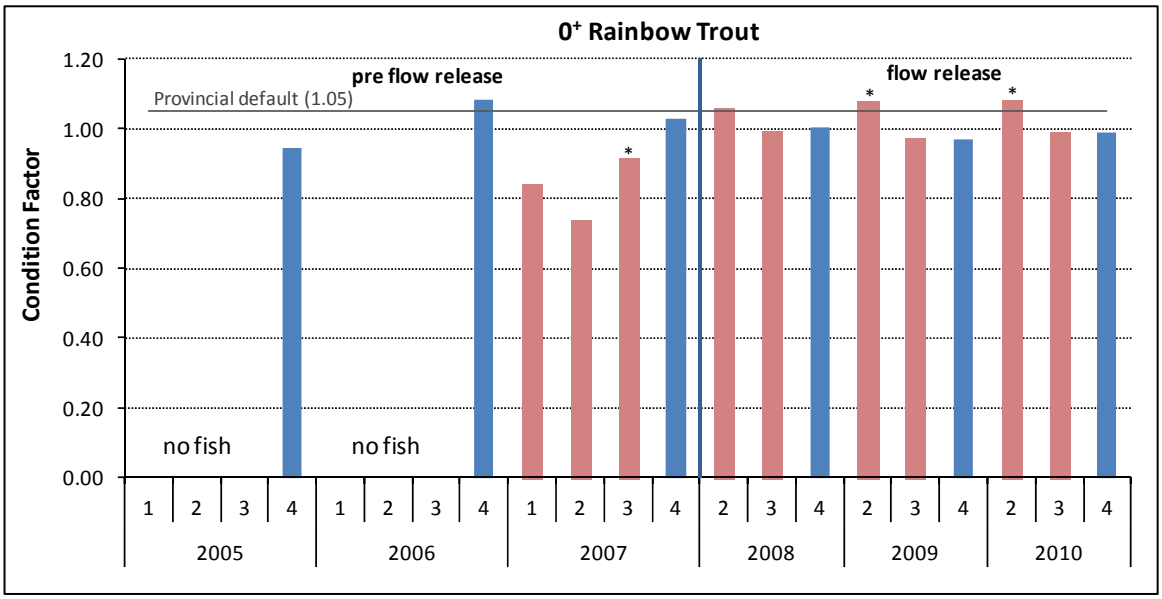

Figure 13. Mean condition factor of rainbow trout fry (age 0+) by site and study year from JORMON#2 (based on data from Burt 2011 (draft)). Bars with an asterisk are significantly different than Site 4 ($p < 0.05$, Dunnett's Test with Site 4 taken as the control). Note the poor condition factor at Sites 1, 2, and 3 in 2007.

Information presented in this section provide some clues as to the threshold level of copper, below which rearing becomes possible in the affected zone of the Jordan River. For example, in 2005 and 2006 copper concentrations did not permit rearing in this zone; in 2007 rearing occurred but was impaired (poor condition factor); in 2008–2010 rearing occurred throughout all sites (Figure 11) and fish displayed a normal condition factor (Figure 13). To get an understanding of copper levels in the river during rearing period in each of these study years, the regression equations from Figure 6 were applied against mean monthly discharge data for these years to predict average monthly copper concentrations during the April to October period (Figure 14). Given the performance of rearing prior to 2007, in 2007, and after 2007, one can roughly judge that rearing is not possible when copper levels exceed 0.04 mg/L, become possible but impaired when levels drop below this, and becomes successful with what appear to be healthy fish when levels drop below 0.015 – 0.020 mg/L. A bioassay would be a means to confirm and fine tune the copper threshold for healthy rearing fish.

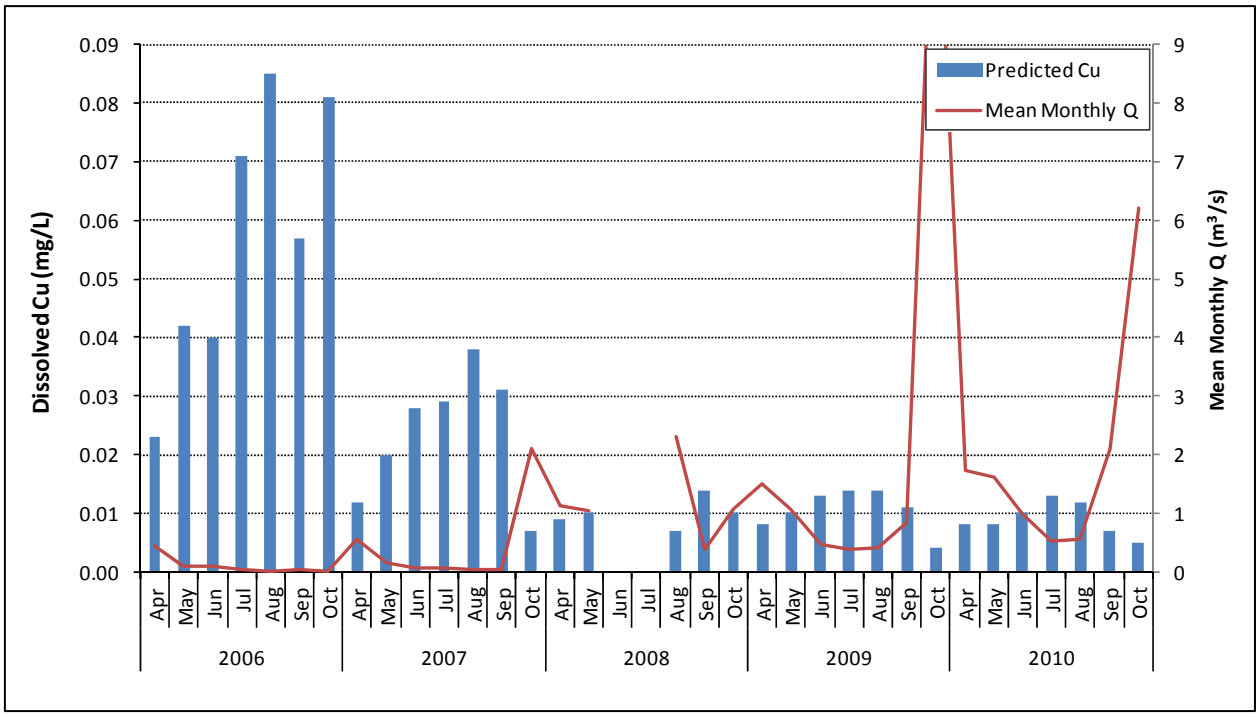

Figure 14. Predicted average monthly dissolved copper concentration in the affected zone for the months of April to October, 2006–2010. Copper concentrations are based the regression equations in Figure 6 applied against mean monthly flows from JORMON#1. Note: values are not presented for 2005 due to limited flow data.

The incubation study by Wright and Guimond (2003) also included a sea water challenge and a copper toxicity bioassay, both using pink fry. The seawater challenge involved placing 50 pink fry from incubation site 3 (Inc 3) into a 75 L bucket containing 27.5 ‰ (parts per thousand) saltwater and holding them for 109 hours. The goal was to determine whether exposure to elevated copper levels during incubation and alevin stages resulted in any adverse consequences in the ability of fry to adjust to the saltwater environment. The results found no mortalities among the 50 pink fry after 109 hours in saltwater indicating that the ability to adapt to saltwater was unaffected by the levels of copper experienced during the incubation and alevin stages.

The bioassay component of their study involved transfer of 630 pink fry from the study's control group (held at the Puntledge Hatchery) to Rosewall Creek Hatchery where they were divided among 30 5-litre buckets containing water from Site 4, spiked with differing concentrations of copper. The assay buckets included 9 different copper concentrations ranging from 28 to 165 µg/L (0.028 – 0.165 mg/L), a control group with no added copper (background copper was 5.5 µg/L), and there were 3 replicates for each of the 10 assay groups. Mean percent survival from each concentration group are shown in Figure 15. Results indicated increased mortality at the 2 highest concentration treatments: 9.9% mortality at 139 µg/L and 20.5% mortality at 165 µg/L. When treatment means were tested for significance against the control mean (Dunnett's Test), only mortality for the 165 µg/L group was

significant (p = 0.02). The best fit to the data was an exponential curve showing increased mortality with increased copper concentration (r^2 = 0.873).

Wright and Guimond (2003) noted that the bioassay failed to identify the LC50 for Jordan River water (LC50 = the "lethal concentration" that kills 50% of the test animals). They noted that these results differed substantially from a similar pink fry bioassay completed with Tsolum River water where the LCC50 was 120 µg/L (0.12 mg/L).

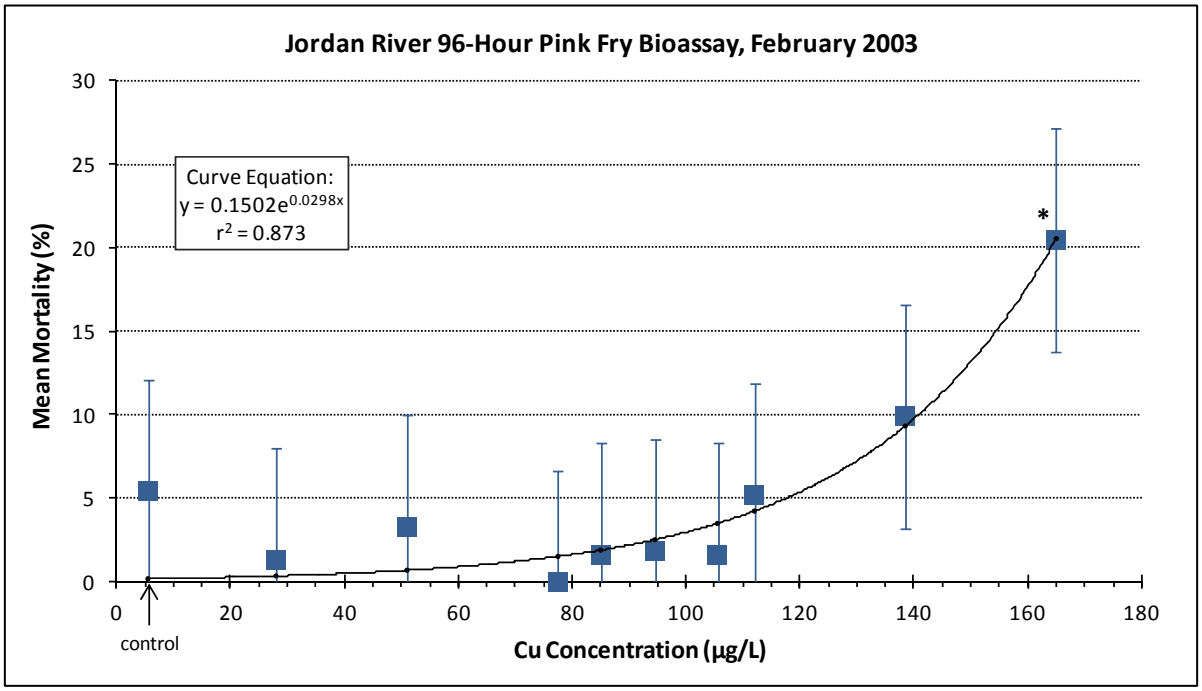

Figure 15. Mean mortality (%) of pink fry exposed to differing concentrations of copper. Means are the average of 3 replicates per concentration group; error bars are 95% confidence limits based on a pooled variance from all groups. Mean mortalities significantly different than the control are indicated by an asterisk. The best fit to the data (highest r^2) was an exponential curve. Data were adapted from Wright and Guimond (2003).

5. DISCUSSION

Water samples from the various Jordan River studies indicated that the copper levels in the impacted zone ranged from 0.007 – 0.086 mg/L for dissolved copper, and from 0.008 – 0.127 mg/L for total copper, and that concentration was dependent on discharge at the time of sampling (Table 2, Figure 6). These values are well above the Provincial criteria for 0.002 mg/L for freshwater aquatic

life (30-day average when hardness ≤ 50 mg/L, Ministry of Environment 1998). Despite the comparatively high readings in the copper affected zone, the biological data suggest no observable impacts to egg and alevin stages for coho and pink salmon. However, for the rearing stages of rainbow trout and coho salmon, there were definitive impacts as indicated by a complete absence of fish at dissolved copper concentrations > 0.04 mg/L. At dissolved copper concentrations between 0.04 and roughly 0.015–0.20 mg/L, rearing occurred but fish exhibited poor condition factors suggesting that health was impaired. The threshold for successful rearing appears to be around 0.015 – 0.020 mg/L, and when dissolved copper was below this level, coho fry and various age classes of rainbow trout were found rearing in the affected zone, and displayed normal condition factors (Figures 11 and 12).

Differences in copper tolerance between egg and rearing stages have been documented in other studies. In their literature review on copper toxicity, Richey and Roseboom (1978) noted various studies that found egg stages to be more tolerant of copper than rearing stages (salmonid examples included brown trout and chinook salmon). Reported effects of elevated copper concentrations on rearing salmonid fry included darkened colour and loss of appetite. Thus the thinness of rainbow fry captured in the affected zone 2007 may have been due to a loss of appetite. An additional possibility, is that copper levels may have adversely affected food supply, though the effect of copper on aquatic invertebrates has not been examined on the Jordan River.

The JORMON#2 Fish Index Study (Burt 2011 (draft)) found that rainbow trout and coho fry successfully reared within the copper affected zone during 2008 to 2010 (after initiation of the fish flow release). Monthly mean dissolved copper during the main rearing months (June to September) for these years was estimated to range from 0.007 to 0.014 mg/L with an overall average of 0.012 mg/L (Figure 14). As previously noted, this is substantially greater than 30-day average of 0.002 mg/L established by the Province for the protection of freshwater aquatic life. Thus, it appears that there are aspects in the chemistry of Jordan River water that are reducing the toxicity of copper to fish.

Literature on factors influencing copper toxicity indicated that dissolved organic carbon (DOC), pH, alkalinity (buffering capacity), and hardness (level of calcium ions) are important features of water chemistry influencing copper toxicity (McIntyre et al. 2008, Richey and Roseboom 1978, Solomon 2009). Specifically, toxicity is reduced with higher levels of these parameters. DOC is reported to be the most important factor due to the fact that copper forms strong complexes with DOC thereby reducing the amount that is bioavailable. pH plays a role in this complexing in that low pH causes the bonds to break down resulting in more copper being bioavailable. Waters with high buffering capacity (alkalinity) also reduce bioavailable copper through complexing with carbonate and bicarbonate ions. Lastly, waters with higher hardness have greater concentrations of calcium ions, which offer a protective effect on cells.

Table 4 summarizes values for these parameters in the copper affected zone of the Jordan River. None of the water chemistry analyses performed on Jordan River samples included DOC, however,

in the Provinces guide for DOC criteria (Moore 1998), it is reported that most BC streams have organic carbon concentrations of < 5 mg/L, the exceptions being waters with high natural sources such as bogs, marshes, and wetlands. The other parameters were available from the various studies conducted on the river. It is important to recognize that alkalinity and hardness are inversely relate to discharge and so the most telling samples are those collected during summer base flow. These data show a number of water quality features in favour of DOC complexing, in particular the relatively high pH (7.2), and the notable increase in alkalinity after initiation of the fish flow release (from 11.5 to 18 mg/L). As mentioned in the Background section, this is thought to be due higher alkalinity in the fish flow release water which forms a major portion of downstream flows during summer low flows.

Table 4. Summary of Jordan River data for water chemistry parameters known to influence copper toxicity. Values are from sampling in the copper impacted zone of the Jordan River.

Water Chemistry Parameter	Values in Copper Zone	Source
DOC (mg/L)	< 5	Typical range for BC streams (Moore 1998)
pH (pH units)	7.2	6-year mean from Burt (2011 (draft))
Alkalinity (mg/L)	11.5 before/18 after	3-year means from Burt (2011 (draft))
Hardness (mg/L)	15	August 19, 1994 value from Griffith (1996)
Water temperature (°C)	14.9	Aug. mean (2006, 2008-2010) from the JORMON#2 temperature logger located at the lower river gauging station (Burt 2011 (draft))

In this report, the assessment of the effects of copper on Jordan River salmonids has focused on observable effects — percent survival of eggs during incubation, fish presence or absence during rearing, condition factor of rearing fish, etc. However, the literature indicate that exposure to copper may produce less tangible adverse effects. For example, laboratory studies have shown that coho fry exposed to copper experience impairment of their olfactory function (sense of smell), and that the degree of impairment was proportional to the concentration of copper (McIntyre et al. 2008, Sandahl et al. 2004). Disruption mechanisms included copper ions binding with olfactory receptor sites making them unavailable for natural odorant molecules, and interference of neuron receptors and intracellular pathways responsive to different natural odorants. The significance of these findings is potentially quite broad since olfactory function is required for locating food, avoidance of predators, home stream imprinting, adult homing, and reproduction. Thus, though these copper induced impacts may not direct cause mortality, they may eventually cause indirect mortality.

Another effect that is difficult to observe is that exposure to copper may elicit a physiological stress response in rearing salmonids. Taub (2004, cited in Solomon 2009) found that exposing rainbow trout to low levels of copper caused hyperactivity, increased blood levels of the stress hormone cortisol, and synthesis of the metal-detoxifying protein metallothionein in the liver. These physiological responses are known to weaken immune systems and could increase susceptibility to disease.

6. CONCLUSIONS

The following are key points drawn from this assessment of available water quality and biological data for the anadromous reaches of the Jordan River:

- Water samples collected on the anadromous reaches of the Jordan River (1994 to 2012) show an abrupt increase in copper concentration beginning adjacent to the large deposit of material left on the left bank by historic mining activities. Dissolved copper data from within the impacted zone ranged from 0.007 – 0.086 mg/L (Table 2), were inversely related to stream discharge, and fit well to a power curve equation (Figure 6). In contrast, copper levels upstream of the mine deposit ranged from <0.001 to 0.004 mg/L and did not vary with stream discharge.

- Most studies conducted on the lower Jordan River have assumed that the mine deposit on the left bank is composed of copper "tailings", and that this is the source of elevated copper in the lower river. However, the inspection of this deposit by Burke (2012a) indicated that the particle size and colour are inconsistent with tailings, and that the deposit may in fact be excavations from creation of the 2.6 km access tunnel in 1956. Thus, there remains uncertainty as to the actual source of the copper.

- Water quality monitoring by the JORMON#2 study suggests that copper levels within the impacted zone decreased incrementally, first in 2007, and then again in 2008. The 2007 decrease was believed to be due record floods during the 2006/07 winter, which scoured the channel and washed the toe of the mine deposit out to the ocean. The second decrease occurred following initiation of fish flow releases at Elliott Dam in January 2008. This second decrease in copper concentration was due to the dilution effect provided by these flow releases. Due to concerns of the fish release control valve tripping, the valve has been maintained in a locked open position. In this position, flow releases at Elliott Dam typically range from 0.30 – 0.40 m^3/s as opposed to 0.25 m^3/s prescribed in the Jordan River WUP.

- Biological studies conducted on the Jordan River provide information on salmonid spawning, incubation, and rearing within and upstream of the copper affected zone. The spawning surveys were conducted from fall 2005 to spring 2011 and involved snorkel enumerations during coho and steelhead spawning periods. Adult numbers during the 6 survey years were very low (Figure 8), but when fish were found, it was frequently in the riffles adjacent to the mine deposit. This

suggests that adults did not avoid the copper affected zone. Of note is that flows during spawning periods tend to be higher than in the summer, which would have diluted copper inputs.

- Incubations studies using coho or pink salmon eggs were conducted from 2000 – 2011. Results generally indicated egg-to-emergence survival rates of ≥ 90%, with no apparent difference between sites within the impacted zone and those above this zone (Figure 10). Sites with poor survival were attributed to siltation, and/or low intergravel flow/dissolve oxygen conditions as opposed to copper toxicity. Thus, these studies suggest that incubation success is not diminished by ambient levels of copper in the impacted zone.

- Rearing studies included a 1994 inventory by Griffith (1996) and the 6-year WUP monitor (JORMON#2), with both investigations employing similar enumeration techniques. Results indicated a total absence of rearing fish within the impacted zone until 2007, when rainbow fry were encountered, albeit with extremely low condition factors. The presence of fry in this zone in 2007 was attributed to a reduction in copper due to the previous winter's floods as previously described. However, the thinness of the fry suggested copper levels were still causing adverse effects on fish health. In January 2008, the fish flow commenced, and subsequent inventories from 2008 to 2010 captured both rainbow fry and parr within the impacted zone, and at densities similar to the site upstream of the copper zone (Figure 11). Furthermore, fish captured in the impacted zone during 2008–2010 showed condition factors similar to fish at the upstream site (Figure 13). The occurrence of multiple age classes, of similar densities and condition factor to the control site suggests that copper levels may be sufficiently reduced under the current flow release regime to permit successful rearing of rainbow trout.

- The performance of rearing prior 2007, in 2007, and after 2007 provided an opportunity to assess the threshold level of copper below which rearing becomes successful in the impacted zone. For this assessment, the copper-discharge regression equation from Figure 6 was applied against mean monthly flows during rearing months to estimate mean monthly copper concentrations during 2006 to 2010 (Figure 14). Results suggested that rearing is successful when dissolved copper concentrations are less than ~ 0.015 – 0.020 mg/L. From this level to about 0.040 mg/L, rearing is possible but is impaired (sublethal impacts). When dissolved copper levels exceed 0.40 mg/L the data suggest rearing is not possible.

In conclusion, studies conducted on the Jordan River suggest that under the current flow releases from Elliott Dam (0.3 – 0.4 m^3/s), copper inputs into the lower river are sufficiently diluted to allow for successful completion of spawning, incubation, and rearing. The data indicate that the rearing stage is the most sensitive to copper, but it appears that as long as the concentration of copper remain below ~0.015 – 0.020 mg/L, rearing can occur and fish are able to feed and grow normally (as indicated by condition factor). The one uncertainty in this conclusion is whether there are more subtle adverse effects occurring that are not detectable by available studies on the river. For example, it is difficult to ascertain whether some level of olfactory impairment is occurring among rearing fish,

though intuitively, if this were severe one would not expect rearing fish to display a normal condition factor as occurred during 2008 – 2010. Similarly, existing studies cannot detect whether fish rearing in the affected zone have elevated stress hormones (cortisol) or are synthesizing detoxifying proteins in their livers.

7. RECOMMENDATIONS

The most important task is to determine with certainty the source of copper contamination. This should involve a reconnaissance field trip to inspect the mine deposit and area around the portal. Activities should include:

- Collect soil samples of the bank deposit for composition analysis.
- Explore the area along the bank deposit and around the portal for all potential sources of copper seepage.
- Collect water samples at all seepages found for subsequent lab analysis of dissolved and total copper.
- Though it would not take the place of water samples, it may useful to test various seepage sources using a copper field test kit (detection limits are typically ~ 0.5 mg/L). This would provide a means to quickly assess for high concentration sources.

The value of this work is that it is not possible to undertake remedial work to cut off or eliminate copper inputs to the river until the source is known. For example, a lot of money could be spent to remove the bank deposit but this would not solve the problem if it is not the source of the copper. Alternatively, if the source turns out to be the access tunnel, then efforts and funds could be directed towards this location instead of the bank.

Once the source of copper is determined, then it is further recommended some research time be spent looking into the literature as to possible approaches for eliminating copper inputs to the Jordan River. This could possibly involve consulting an expert in mine cleanup.

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